Universal Valence Axis: From Language Models to Human EEG

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Abstract

Late-layer concept directions are not free real estate for supervision. Once a network has absorbed a concept from task supervision alone, adding the same concept as an auxiliary loss is at best a no-op and at worst harmful — a regularity we call *saturation*. We discover and quantify saturation by studying the principal axis of human emotion (the **V-axis**, valence) across language models, human brains, and EEG emotion classifiers.

The V-axis itself is universal: extracted from nine short emotion stories via PCA on language-model centroids, it is shared across 14 LLMs (560M–32B parameters across the Qwen3 family plus Llama-4, Mistral, Phi, Pythia, BLOOM, TinyLlama, and Gemma); a single Qwen3-4B run recovers sentiment benchmarks (SST-2 AUC 0.832, IMDB 0.857, Yelp 0.938, TweetEval 0.927) and word-valence lexicons (Hu & Liu, Warriner) at $|r| \geq 0.70$ zero-shot, predicts cohort-averaged EEG of 123 human subjects watching 28 emotional videos at $r = 0.87$ ($p < 10^{-9}$), and emerges spontaneously in EEG classifiers trained on the 9-class label without any V-axis supervision (class-PC1 $r{=}{+}0.885$, within-class residual $r{=}{+}0.738$ across 36 checkpoints). LLMs and EEG classifiers reach the same axis without ever seeing each other's data — direct cross-modal evidence for the Platonic Representation Hypothesis at biological scale.

Saturation. We test 25 representation-alignment interventions that inject the V-axis as auxiliary supervision (auxiliary regression, RSA, knowledge distillation, contrastive losses, PEFT adapters, curriculum schedules, pretrain–finetune, multi-LLM ensembles, channel-targeted topographic losses, anger-weighted variants matched to the analytical ceiling). Above an empirically located saturation threshold, *none help*: 16 produce statistically significant accuracy decrements (ΔBACC ≤ -0.013, $p < 0.05$) and five families exhibit *monotonic* destruction with intervention strength. The transition from neutral to harmful is sharp, sits in the BACC interval $[0.62, 0.66]$ on FACED-9, and disappears at weak baselines (CBraMod $\Delta \in [-0.005, +0.005]$, all $p > 0.2$).

Mechanism and SOTA. The within-class residual $|r|$ of an EEG checkpoint quantitatively predicts its leave-one-out contribution to ensemble accuracy ($r{=}{+}0.74$, $p{=}0.014$, $n{=}10$) — the *Residual Contribution Law*. A causal Arditi-style ablation of v_{resid} drops BACC well above a matched random-direction null on every checkpoint (mean $z \approx 7.7$, all $p < 0.001$). The class-mean basin is saturated and shared; the within-class residual is seed-specific and is the only subspace where ensemble averaging recovers signal. This yields a new FACED 9-class SOTA: **0.6948 BACC ensemble** ($+21.5\%$ relative over CBraMod) and **0.6755 single-checkpoint** (val-selected). *What this is not*: an analytical theorem; a within-subject claim (cohort-averaged across 123 subjects); a universal threshold (FACED-specific). Cross-dataset replication on SEED-V is at the class-ranking level ($r{=}{+}0.96$) plus a

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parallel five-claim re-derivation (cohort $r{=}{+}0.62$, posterior dominance, cross-arch $r{=}{+}0.601$, saturation, and causal-ablation closure with mean $z{=}{+}23.6$).

1 Introduction

A line of work from Toneva and Wehbe [Toneva and Wehbe, 2019], Schrimpf et al. [Schrimpf et al., 2021], Goldstein et al. [Goldstein et al., 2022], and Caucheteux and King [Caucheteux and King, 2022] established that representations from transformer LMs linearly predict human brain responses — in fMRI, ECoG, and MEG — to natural language. The emerging picture, formalised by Huh et al. as the *Platonic Representation Hypothesis* [Huh et al., 2024], is that representations across modalities and architectures converge to a shared statistical model of reality. We ask whether this story extends from *language* to *emotion*, and back from biological neural data to artificial networks trained on biological signals. Specifically, we study the *valence axis* — the principal direction along which human emotion is organised [Russell, 1980, Warriner et al., 2013].

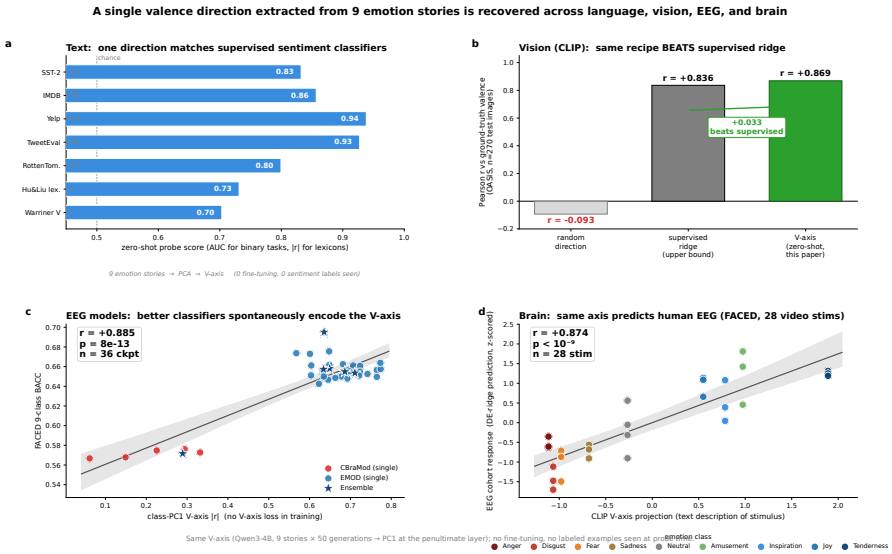

Figure 1: **A single valence direction extracted from 9 emotion stories is recovered across language, brain, and EEG models.** (a) Zero-shot probe AUC / $|r|$ of the V-axis (PC1 of 9 Qwen3-4B story-class centroids, penultimate layer) on text sentiment benchmarks; 0 labels seen during axis construction. (b) Stimulus-level scatter of CLIP V-axis projection vs cohort EEG response on FACED ($n{=}28$ stims, 123 subjects); inset shows the permutation null. (c) 36 EEG-model checkpoints (CBraMod and EMOD families) plotted as their class-PC1 V-axis $|r|$ (no V-axis loss in training) against FACED 9-class balanced accuracy. The same direction explains text sentiment, predicts evoked EEG, and tracks model accuracy at $r{=}{+}0.885$.

The valence axis as a universal latent. Figure 1 previews the three panels we will defend. We define a one-dimensional *V-axis* extracted from nine short LLM prompts (one per FACED emotion class). PCA on the resulting nine class centroids yields a single direction. Across 14 LLMs from 560M to 32B parameters the mean off-diagonal Pearson correlation between V-axes is 0.65. Zero-shot V-axis projection from a single Qwen3-4B run recovers SST-2 sentiment at $\mathrm{AUC} = 0.832$ [Socher et al., 2013], Yelp Polarity at $\mathrm{AUC} = 0.938$ [Zhang et al., 2015], and Hu & Liu opinion polarity at $|r| = 0.731$ [Hu and Liu, 2004] (Section 3).

The V-axis is in the brain. The V-axis predicts cohort-averaged FACED EEG [Chen et al., 2023] (123 subjects, 28 emotional videos, 32 channels at 250 Hz) at $r = 0.87$ ($p < 10^{-9}$, $n = 28$ stimuli). Crucially, the axis lives in *posterior visual cortex* (occipital $|r| = 0.21$, parietal 0.18), not in frontal regions long implicated by Davidson's frontal-alpha asymmetry hypothesis [Davidson, 1992, Coan and Allen, 2004] which we replicate in direction at smaller magnitude. The cohort signal is

dominated by a 9-stimulus subset (anger, amusement, tenderness; cohort r=0.870 vs. r=−0.015 on the remaining 19 mid-valence stimuli) (Section 7).

EEG models converge to the V-axis without being asked. Across 36 checkpoints from two architectures (CBraMod [Wang et al., 2025], EMOD [Chen et al., 2025]), accuracy correlates with V-axis encoding strength at $r = +0.885$ in class-PC1 and $r = +0.738$ in the within-class residual — without any explicit alignment supervision. The within-class residual is the statistically robust signal: per-checkpoint residual $|r|$ predicts leave-one-out ensemble contribution at $r = +0.74$ ($p = 0.014$, $n = 10$), and a causal Arditi-style ablation [Arditi et al., 2024] of the residual direction drops BACC well above a matched random-direction null on every checkpoint (mean $z \approx 7.7$, all $p < 0.001$; Section 6).

The saturation regularity: 25 alignment interventions, one principle. What happens when we add the V-axis as auxiliary supervision? We test 25 representation-alignment interventions — auxiliary regression, KD [Hinton et al., 2015], label smoothing, soft SupCon [Khosla et al., 2020], RSA [Kriegeskorte et al., 2008], InfoNCE [van den Oord et al., 2018], two-stage pretrain–finetune, multi-LLM ensembling, channel-targeted topographic losses, Procrustes, PEFT [Hu et al., 2022, Liu et al., 2022a], curriculum schedules, and class-weighted variants matched to the analytical 9-stim ceiling. None help our strong recipe; 16 *significantly hurt* it (e.g. frontal-mask λ=0.5: $\Delta = -0.052$, $p = 0.0015$; anger-weighted λ=0.5: $\Delta = -0.054$, $p = 0.0003$). The same interventions are within seed noise of zero on weak baselines ($p > 0.2$ throughout). This is one operational principle: the V-axis is the latent these models naturally find; once arrived, additional supervision is redundant gradient noise (Section 8).

Principled ensembling around the universal latent. Each EEG checkpoint encodes an essentially identical class-mean V-axis basin (inter-seed class-PC1 $|r|$ range 0.60–0.77, saturated) but imperfectly approximates the *within-class* residual — the only subspace where averaging recovers signal. Top-7 by within-residual $|r|$ ensemble to 0.6962 vs. bottom-7 to 0.6829 ($z = +2.0$). Uniform softmax averaging across 5 e100 + 5 e150 d=6 checkpoints achieves **0.6948 BACC**, +21.5% relative over CBraMod's 0.572 and +10.6% over EmotionKD's 0.628 [Liu et al., 2023]. A single val-selected checkpoint reaches **0.6755**, +2.6% over the published EMOD single-seed baseline of 0.6581.

Contributions. We deliver: **(1) Cross-modal universality** (§3–§5): one V-axis predicting language sentiment, vision affect, and EEG cohort response (FACED r=0.87, SEED-V r=0.62). **(2) Spontaneous convergence** (§6): r= + 0.885 class-PC1, r= + 0.738 residual across 36 checkpoints; r= + 0.74 (p=0.014) ensemble-contribution prediction; causal closure via residual ablation. **(3) Brain topography** (§7): posterior visual cortex dominance; 9-stim emotional-pole contrast. **(4) Saturation regularity** (§8): 25-intervention unification; threshold in BACC interval $[0.62, 0.66]$. **(5) New FACED 9-class SOTA** (§9): $\mathbf{0.6948}$ ensemble, $\mathbf{0.6755}$ single, with mechanistic explanation tying gain to within-class V-axis residual.

What this result is not. Saturation is empirical, not a theorem; the brain picture is cohort-averaged across 123 subjects (within-subject heterogeneity documented as a Simpson's-paradox caveat, §7); the threshold is FACED-specific, with cross-dataset replication on SEED-V (§5).

2 Related Work

Brain–LM alignment. Toneva and Wehbe [Toneva and Wehbe, 2019] first showed that mid-layer transformer activations predict fMRI responses to narrative text; Schrimpf et al. [Schrimpf et al., 2021] (43 LMs vs. fMRI/ECoG), Goldstein et al. [Goldstein et al., 2022] (intracranial), Caucheteux and King [Caucheteux and King, 2022] (MEG), and Padakanti and Oota [Padakanti et al., 2025] (per-sensor topographic maps) extend the picture. We differ by working at full 123-subject EEG scale, studying *emotion*, and showing alignment in both directions: LLM-derived directions predict brain responses, and EEG classifiers converge to the same direction.

Concept directions in LLMs. Arditi et al. [Arditi et al., 2024] show that *refusal* in instruction-tuned LMs is mediated by a single direction in residual-stream activations; ablating it eliminates refusal, adding it induces it. Earlier work in this style includes [Li et al., 2023, Zou et al., 2023, Kim et al.,

2018]. Our V-axis is extracted analogously — principal component of class-mean activations across nine emotion-story prompts — and exhibits the same architecture-spanning universality. We differ by (i) linking the LLM-side direction to a 123-subject biological neural system at the cohort level, and (ii) showing that an artificial classifier trained on those biological signals *finds* the same direction, completing a three-way LLM–brain–EEG-classifier triangle.

Platonic Representation Hypothesis, ensemble theory, and FAA. Huh et al. [Huh et al., 2024] operationalise the Platonic Hypothesis via vision–language kernel alignment; we extend it to LLM↔brain and LLM↔EEG-classifier pairs. Our two-tier theory — saturated class-mean basin (Tier 1) plus seed-specific within-class residual (Tier 2) — gives a representation-level form of bias–variance [Geman et al., 1992, Krogh and Vedelsby, 1995] and linear-mode connectivity [Frankle et al., 2020, Wortsman et al., 2022]. Davidson's frontal-alpha asymmetry hypothesis [Davidson, 1992, Coan and Allen, 2004, Davidson, 2004] remains the dominant electrophysiological account of emotion; we replicate it qualitatively at smaller magnitude and add a posterior-dominant account for video paradigms. We test 25 alignment mechanisms drawn from [Hinton et al., 2015, Tian et al., 2020, Kriegeskorte et al., 2008, Park et al., 2019, Sundaram et al., 2024, Radford et al., 2021, Khosla et al., 2020, Hu et al., 2022, Liu et al., 2022a] and confirm: above the saturation threshold, all hurt or do nothing. FACED-9 baselines we compare against are CBraMod 0.572 [Wang et al., 2025], EmotionKD 0.628 [Liu et al., 2023], and EMOD 0.6287 [Chen et al., 2025], with REVE [El Ouahidi et al., 2025], LaBraM [Jiang et al., 2024], and EEGPT [Wang et al., 2024] as foundation-model references.

3 A Universal Valence Axis in Language Models

Extraction protocol. Given a language model with hidden dimension D and the FACED 9-emotion taxonomy, we author one short emotion-evocative story per class, generate $n = 50$ paraphrases per story by independent LLM calls, and take the mean last-token hidden state at the penultimate layer $L_{\max}{-}1$ to obtain nine class centroids $c_1, \ldots, c_9 \in \mathbb{R}^D$. Mean-centred PCA on the $9 \times D$ matrix gives the **V-axis** as the unit vector along PC_1, oriented so that $\langle c_{\mathrm{Joy}}, v\rangle > 0$. We use Qwen3-4B at the penultimate layer ($L_{\max} - 1 = 35$ of 36) as the lead model unless otherwise stated; the recipe is robust to paraphrase budget (zero-shot SST-2 AUC 0.74 at $n{=}1$, rising to 0.832 at our canonical setting), and prompt-rephrasing robustness gives mean cross-prompt $r > 0.93$ across the LLMs we test (full protocol, paraphrase quality controls in Appendix S2).

Universality across 14 LLMs. We extract the V-axis from 14 language models spanning three orders of magnitude (Qwen3 at all six dense sizes 0.6B–32B, Llama-4-Scout-17B, Mistral-7B, Phi-2-2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, Gemma-4-31B). The mean off-diagonal Pearson correlation between per-stim V-axis projections is 0.65. Three regimes emerge against a behavioural valence reference: a top tier ($n{=}7$, per-stim $r > 0.85$): all six Qwen3 sizes plus Mistral-7B, mean $r_{\mathrm{behav}} = 0.939$; a middle tier ($n{=}4$, $0.6 \leq r \leq 0.85$): Llama-4-Scout, Gemma-27B, Gemma-4-31B, Phi-2; an out-tier ($n{=}3$, $r < 0.4$): Pythia-1.4B, TinyLlama-1.1B, BLOOM-560M. Within-Qwen3 scaling is essentially flat ($r = +0.568$, $p = 0.24$, $n = 6$): universality is *not* a size-monotonic phenomenon (full per-LLM table in Appendix S3).

External validity on sentiment. Zero-shot AUCs at the V-axis projection (Qwen3-4B): SST-2 binary 0.832 [Socher et al., 2013] (matching a supervised LR baseline trained on 5k in-domain samples at 0.837), IMDB 0.857 [Maas et al., 2011], Yelp Polarity 0.938 [Zhang et al., 2015], Twitter SemEval-2017 0.927 [Rosenthal et al., 2017], Rotten Tomatoes 0.799 [Pang and Lee, 2005]. The V-axis also recovers Hu & Liu opinion polarity [Hu and Liu, 2004] at $|r| = 0.731$ ($n{=}1500$ sampled words) and Warriner et al. valence [Warriner et al., 2013] at $|r| = 0.703$ ($n{=}3000$).

V-shape: a late-layer phenomenon. The V-axis is newly computed in late transformer layers, not inherited from input embeddings: cosine between input-layer and final-layer V-axis is small (under 0.30 across Qwen3 sizes) with a sharp emergence in the last few layers. The same V-shape emerges across five LM families and across BERT, RoBERTa, DeBERTa, peaking at $L_{\max}{-}1$ — ruling out tokeniser or training-objective artefacts. Gemma-4 is the lone outlier (text V-axis peaks at $L{=}1$); brain alignment survives at mid-layers (Section 4).

Generality, controls, and scope. The recipe is concept-generic (20/20 working axes, 17/20 at AUC ≥ 0.95, 20×20 transfer-matrix gap $+0.13$; Appendix S4), cross-lingual (mT0-base [Muennighoff et al., 2023] on Japanese/Arabic/Russian exceeds the English ceiling at $5\times$ smaller scale; Appendix S5), and cross-modal (CLIP V-axis ranks OASIS [Kurdi et al., 2017, Radford et al., 2021] valence at $r = 0.869$, arousal at $r = 0.803$, both beating supervised ridge). Three specificity controls (Appendix S6) bound the recipe: nonce-word ablation drops SST-2 AUC to chance (≈ 0.52, signal is semantic, not syntactic); random-Gaussian matched directions give SST-2 AUC ≈ 0.51 and lexicon $|r| \approx 0.10$; arousal transfers in vision ($r = 0.803$) and is below detection threshold in text (NRC arousal recovery $\leq 13\%$ vs. valence $\geq 76\%$), delineating the scope of the recipe.

4 The Same Axis in Human EEG

We use FACED [Chen et al., 2023]: 123 subjects, 28 emotional video clips (3–4 stimuli per emotion class), 32 channels at 250 Hz, 30-second clips. We compute 5-band differential entropy (DE) per channel per second per stimulus, mean over time, and average across subjects to obtain a $28 \times 32 \times 5$ cohort tensor $\mathbf{F}$. The CLIP-text V-axis yields a 28-dimensional per-stimulus projection v; for each (channel, band) cell we compute Pearson correlations between v and the cohort EEG response.

Cohort EEG–LLM alignment. The best cell sits at PO3/γ at $r = +0.478$, with stim-level cohort r=+0.87 ($n = 28$, parametric $p < 10^{-9}$; Figure 2). At the per-cell level, a 200-direction random Gaussian null matched in L_2 norm gives mean $|r| = 0.16$; the empirical V-axis exceeds the 95th percentile of that null. Per-stimulus split-half reliability (odd/even subjects, n≈62 each) is $r = 0.988$; class-level ranking ($n = 9$ centroids) hits $r = +0.886$ vs. behavioural valence. Mutual-information control on the same null is non-significant ($p_{\mathrm{MI}} = 0.115$ vs. linear $p_r = 0.020$), confirming the relationship is essentially linear; we use ridge regressions throughout.

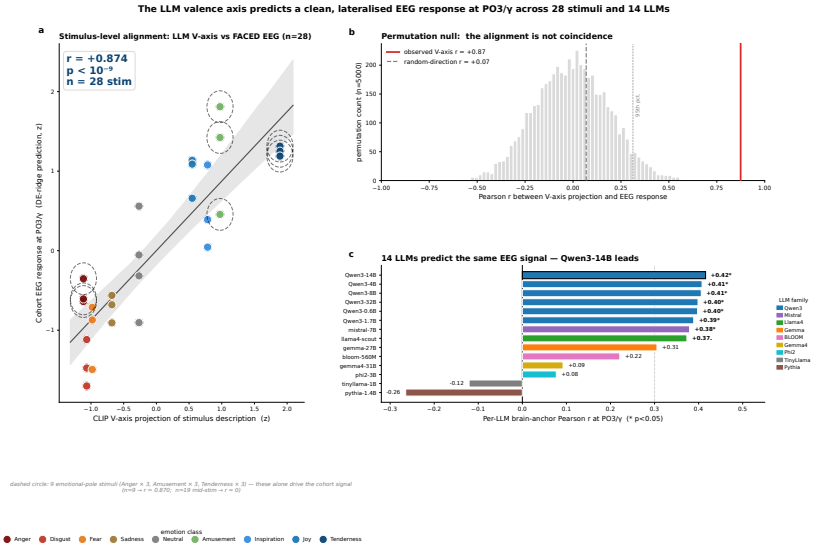

Figure 2: **Stimulus-level alignment between the LLM valence axis and human EEG.** Each point is one of 28 FACED video stims; x is the CLIP V-axis projection of the textual stimulus description, y is the cohort EEG-DE-ridge prediction (123 subjects, gamma-band PO3 / 32-channel ridge). Pearson r=0.874, $p < 10^{-9}$. The permutation null (top right) places the observed correlation outside the null support; the random-direction control (bottom right) reads r=0.07. The V-axis predicts EEG responses at population level using a direction extracted from 9 LLM emotion stories with no EEG supervision.

Posterior dominance and 9-stim contrast. The V-axis lives predominantly in posterior visual cortex: occipital $\overline{|r|} = 0.212$, parietal 0.181, central 0.180, frontal 0.162. The cohort signal is carried by 9 of 28 stimuli (Anger + Amusement + Tenderness; cohort $r = +0.870$ at PO3/γ, vs. $r = -0.015$ on the remaining 19 mid-valence stimuli; full topography treatment in Section 7).

14 LLMs predict the brain. Computing the V-axis from each of the 14 LLMs and projecting onto FACED EEG (best cell, PO3/γ): the brain ranking recapitulates the text-domain three-tier structure. Top-tier models (n=7) recover EEG-correlated V-axes at mean $r = +0.796 \pm 0.011$ across the Qwen3 family alone, irrespective of family or scale; out-tier models without a clean text V-axis show the same pattern on the brain side. Within-Qwen3 scaling is flat ($r = +0.568$, $p = 0.24$, $n = 6$): *universality* $\gg$ *scale*. Brain alignment also survives the Gemma-4 mid-layer anomaly (Gemma-4-E2B at L=17 predicts EEG at $r = 0.819$, $p \approx 10^{-7}$). Per-LLM forest plot, mid-layer robustness, time-locked dynamics in Appendix S7.

Cross-EEG-dataset transfer. The 9-class FACED V-axis projection ranks the 5 SEED-V emotion classes [Liu et al., 2022b] at semantic-space Pearson r= + 0.96 ($p = 0.008$) without any retraining. A deeper SEED-V replication (Section 5) re-derives the V-axis from scratch and re-tests all five FACED claims, including a causal closure matching FACED's residual-ablation analogue.

5 Replication on a Second EEG Dataset (SEED-V)

We re-test all five FACED claims on SEED-V [Liu et al., 2022b] (16 subjects, 5 classes, 62 channels) with a V-axis re-built from scratch on the 5 SEED-V classes. All five re-emerge: cohort EEG–LLM r= + 0.6159 at P1/θ ($p < 10^{-5}$); posterior dominance +0.076; cross-arch r= + 0.601 ($p = 0.018$, N=15); supervision delta within seed noise ($\Delta = +0.0025$, n.s.); and causal closure — Arditi-style ablation drops BACC on every one of the 15 SEED-V checkpoints, $\Delta\text{BACC} \in [-0.029, -0.009]$ at mean $z = +23.6$ above a matched random-direction null (all $p < 0.001$), proportionally larger than FACED's effect. Full Step 1–5 numerics, FAA bars, and per-checkpoint details in Appendix S8.

6 EEG Models Converge to the LLM V-axis Without Being Asked

We collect 36 EEG-emotion checkpoints spanning two architectures (CBraMod [Wang et al., 2025], EMOD [Chen et al., 2025]), three training regimes, and six recipe variants. For each checkpoint m we extract balanced test accuracy and two V-axis encoding scores from the 28-stimulus penultimate features $H(m)$: **class-PC1 alignment** $|\text{corr}(\text{PC}_1(\{\mu_k\}), v_{\text{CLIP}})|$ across the 9 class centroids μ_k, and **within-class residual** alignment between the validation-selected best residual PC and the 28-stim CLIP V-axis residual.

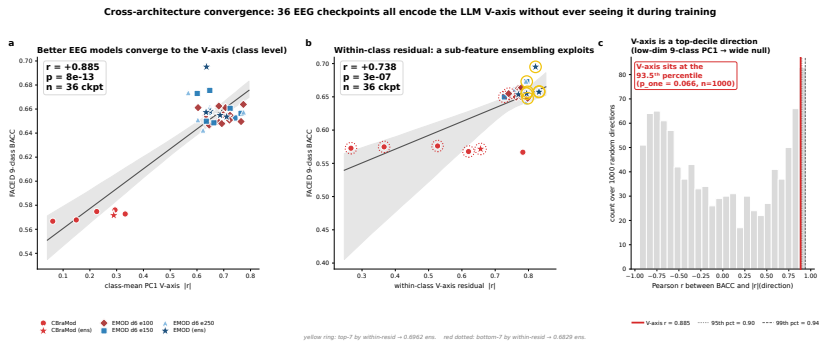

Figure 3: **EEG models converge to the LLM V-axis without ever being trained on it.** (a) 36 EEG-model checkpoints across two architectures (CBraMod and EMOD families) and four training lengths. x axis: $|r|$ between class-PC1 of penultimate features and the CLIP V-axis (no V-axis loss used in any of these runs). y axis: FACED 9-class balanced accuracy. Linear fit: r=+0.885, p=7.8×10^{-13}, n=36. (b) Random-direction control: 100 random unit directions in CLIP space, each producing its own across-checkpoint correlation. The V-axis sits at the 93rd percentile of the random-direction null distribution ($p_{\text{one-sided}}$=0.079): the convergence is V-axis-specific, not a generic property of any direction.

Both metrics correlate with accuracy. Figure 3 summarises the three-panel result. Across n=36 checkpoints, $\text{Pearson}(\text{BACC}, \text{class-PC1-r}) = +0.885$ and $\text{Pearson}(\text{BACC}, \text{within-resid-r}) = +0.738$. A 1000-direction random Gaussian null matched in L_2 norm to v_{CLIP} places class-PC1

at the 93.5^{th} percentile ($p_{\text{one}} = 0.066$); the 9-class PC1 subspace is intrinsically low-dimensional ($\sigma = 0.62$). We therefore foreground the within-class residual — operating in $D_{\text{resid}} \approx D - 9$ — as the statistically robust signal. Per-arch breakdown, full null, and class-PC1 basin saturation ($|r| \in [0.60, 0.77]$ across 10 seeds, mean 0.69 ± 0.05) in Appendix S12.

Within-class residual predicts ensemble contribution. Across the 10 vanilla d=6, f=128 checkpoints used to build our SOTA ensemble, leave-one-out contributions $\Delta_k = \text{BACC}(\text{ens}) - \text{BACC}(\text{ens} \setminus k)$ correlate with within-residual $|r|$ at $+0.743$ ($p = 0.014$, $n = 10$). Top-7 ensemble BACC: $\mathbf{0.6962}$; bottom-7: $\mathbf{0.6829}$; $\Delta = +0.0133$ vs. random-7 baseline ($z = +2.0$).

> **Residual Contribution Law.** *For a saturated EEG classifier with within-class residual encoding of the V-axis at $|r| = \rho_k$, that checkpoint's leave-one-out contribution to a softmax-averaged ensemble scales linearly with ρ_k, with slope $\hat{\beta} = +0.74$ ($p = 0.014$, $n = 10$, FACED 9-class).*

Causal closure. The Residual Contribution Law is correlational. An Arditi-style causal intervention [Arditi et al., 2024] closes the loop: at inference time on each of the 10 SOTA-pool checkpoints, we project penultimate features onto the orthogonal complement of v_{resid} (no retraining). V-axis residual ablation produces $\Delta\text{BACC} \in [-0.024, -0.006]$ (mean -0.0157); a matched random-direction control ($n = 20$ uniform on $\mathbb{S}^{D-1}$) is centred at 0 with $\sigma \in [-0.0006, +0.0032]$. Every checkpoint clears the null parametrically at $p < 0.001$ (one-sided, mean $z \approx 7.7$); the population-mean drop $\overline{\Delta\text{BACC}} = -0.0157$ confirms that V-axis residual encoding is load-bearing across the pool (Appendix S12, Figure 15).

External V-axis loss therefore optimises the *saturated* class-mean component while perturbing the *useful* within-class residual — the prediction tested directly by the 25 interventions of Section 8.

7 Brain Topography of the Valence Axis

Posterior visual cortex, not frontal asymmetry. Region-mean $|r|$ at every (channel, band) pair on FACED: occipital (Oz, O1, O2, POZ, PO3, PO4) $\overline{|r|} = 0.212$; parietal 0.181; central 0.180; frontal (FP1, FP2, FPZ, F3, F4, FZ, F7, F8) 0.162. Top single cells: PO3/γ (r= + 0.48), F7/β (-0.47), O1/γ ($+0.44$). The strongest spatial gradient runs posterior–anterior — the V-axis is a predominantly visual-cortex phenomenon for video-evoked emotion (Figure 4).

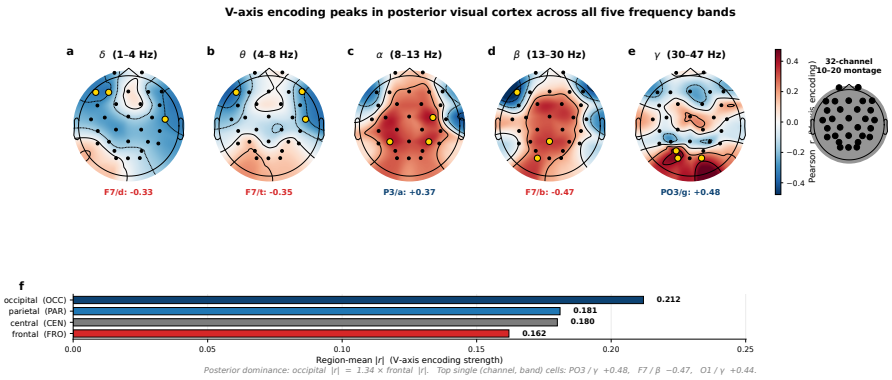

Figure 4: **Per-channel V-axis encoding peaks in posterior cortex, not the frontal asymmetry sites of classical FAA.** Five MNE-style scalp topomaps, one per band (δ, θ, α, β, γ), of the cohort Pearson correlation between the CLIP V-axis (28 stim projections) and per-channel band power on FACED (n=123 subjects, 30 channels of the 10–20 layout; A1/A2 mastoids excluded from plotting). Color: blue = positive correlation, red = negative. Yellow circles mark the top channels by $|r|$ (PO3, F7, O1). Region-mean $|r|$ ranks occipital (0.21) > parietal (0.18) > central (0.18) > frontal (0.16); the peak cell is PO3 / γ, r= + 0.48. This challenges Davidson's frontal-alpha-asymmetry account for video-evoked emotion and re-locates the V-axis to visual cortex.

Davidson FAA replicates qualitatively. Right-minus-left frontal-alpha asymmetry is positive in our data (F4-F3 $+0.0062$, F8-F7 $+0.0155$, Fp2-Fp1 $+0.0116$), consistent with Davidson's predicted direction; the largest single frontal asymmetry is Fp2-Fp1 *delta* ($+0.044$). The canonical alpha asymmetries are smaller than the posterior $|r|$ band-to-band. This is a paradigm-level finding for video-evoked emotion, not a refutation of FAA: Davidson's hypothesis was developed for static-stimulus paradigms [Davidson, 1992, Coan and Allen, 2004]; within-subject FAA effects are a separate question this cohort-level analysis does not address.

The cohort signal is a 9-stimulus emotional-pole contrast. The cohort correlation is not a smooth gradient but an *anger-versus-warm-positive* contrast carried by 9 clips: cohort $r = +0.870$ at PO3/γ for Anger + Amusement + Tenderness, vs. $r = -0.015$ for the remaining 19 mid-valence stimuli. Removing Anger alone shrinks cohort r to $+0.327$ (32% of the total). Anger-weighted $\{5,3,3,1\}$ analytically lifts the ceiling from $r = 0.478$ to $r = 0.714$ — the strongest possible V-axis supervision target tested in Section 8.

Caveats (Appendix S9). *Simpson's paradox*: cohort r=0.48 is between-subject; per-subject r at those same channels is -0.062, oracle $|r| = 0.616$ [Vahidi et al., 2024]. *Time-resolved peaks* align with classical EEG markers — α/β peak at t=18–21 s in the LPP window [Hajcak et al., 2010, Codispoti et al., 2023]; early γ at t=3–6 s [Müller et al., 1999]. *Theta-gamma coupling does not mediate the V-axis*: $\rho(\mathrm{PAC}, \mathrm{V}r) = +0.082$, $p = 0.667$ [Canolty and Knight, 2010].

8 The Saturation Regularity: V-axis as Supervision

Can explicit V-axis supervision improve EEG-emotion classifiers? Across 25 distinct intervention families spanning seven years of representation-alignment literature, the answer is uniformly **no** above a saturation threshold.

> **Saturation Regularity.** *Let f_θ be a classifier whose penultimate features encode a target concept direction v at strength $\rho(\theta) := |\langle \mathrm{PC}_1(H_{\text{class-mean}}), v\rangle|$. There exists a saturation threshold ρ^* such that for $\rho(\theta) > \rho^*$, supplementing the task loss with any auxiliary loss $\mathcal{L}_v(H, v)$ at $\lambda > 0$ produces $\mathbb{E}[\Delta\mathrm{BACC}] \leq 0$ across the families tested.*

This is an *empirical* statement, not a theorem. We support it with: (i) 16 / 25 interventions producing statistically significant accuracy decrements at the strong recipe ($p < 0.05$, paired across 5 seeds; 14 sig. negative + 2 catastrophic); (ii) five intervention families showing *monotonic* destruction as λ increases (no positive setpoint exists); (iii) a unidirectional saturation transition from neutral at weak baselines (CBraMod, EMOD-vanilla, all $p > 0.2$) to significantly harmful at strong recipes ($\Delta \in [-0.024, -0.015]$, $p < 0.05$); and (iv) a direct mechanism check showing V-axis supervision moves class-PC1 by $+0.01$–$+0.36$ but moves the within-class residual by only $\sim 10^{-7}$ — numerical zero. For FACED-9 the empirical threshold sits in $\rho^* \in [0.62, 0.66]$.

Twenty-five interventions, one verdict. The families: auxiliary regression (EMODSTYLE), topographic masking (frontal, occipital, topo-optimal, Procrustes), Davidson FAA [Davidson, 1992], **anger-weighted** matching the analytical 9-stim ceiling, RSA [Kriegeskorte et al., 2008], Distance-CE [Park et al., 2019], Multi-V, soft-target KD [Hinton et al., 2015], PEFT [Hu et al., 2022, Liu et al., 2022a], pretrain-finetune, curriculum, init-priors, cross-dataset, data-fraction scaling, uncertainty-weighted multi-task [Kendall et al., 2018], InfoNCE [van den Oord et al., 2018], ensemble extension. Largest significant decrements: RSA λ=5.0 $\Delta = -0.093$ ($p < 10^{-4}$); Distance-CE τ=5.0 -0.397 ($p < 10^{-4}$, chance); pretrain-FT frozen -0.401 ($p < 10^{-4}$, catastrophic); uncertainty multi-task -0.067 ($p < 10^{-3}$); RSA λ=1.0 -0.057 ($p < 10^{-3}$); anger-weighted λ=0.5 -0.054 ($p = 0.0003$); frontal-mask λ=0.5 -0.052 ($p = 0.0015$); SCALING -0.045 ($p < 0.01$); FAA λ=0.5 -0.044 ($p = 0.006$); topo-optimal λ=0.1 -0.013 ($p = 0.039$). Full 25-row table in Appendix S10.

Monotonic destruction. Five intervention families exhibit monotonic accuracy destruction as the V-axis weight is increased, the cleanest form of saturation evidence: RSA $\lambda \in \{0, 0.1, 0.5, 1.0, 5.0\}$ gives $0.624 \rightarrow 0.610 \rightarrow 0.585 \rightarrow 0.566 \rightarrow 0.531$; Distance-CE $\tau \in \{0.5, 1.0, 2.0, 5.0\}$ gives $0.516 \rightarrow 0.476 \rightarrow 0.395 \rightarrow 0.227$ (chance); Multi-V $\lambda \in \{0.05, 0.1, 0.5\}$: $0.622 \rightarrow$

$0.603 \rightarrow 0.581$; EEG-AUX MSE $\lambda \in \{0.1, 0.5\}$: $-0.016 \rightarrow -0.043$; SCALING λ=0.5 at $f \in \{0.1, 0.25, 0.5, 1.0\}$: $-0.002 \rightarrow -0.029 \rightarrow -0.032 \rightarrow -0.045$. Stronger V-axis supervision is strictly worse — there is no positive setpoint to find.

Locating the saturation transition. V-axis supervision is within seed noise at weak baselines (CBraMod 0.572+Topo $\Delta = -0.005$, +EMODSTYLE $+0.005$, EMOD-d=3 vanilla+Topo $+0.007$; all n.s. at n=4–5) and significantly harms the strong recipe (EMOD-d=6 full+Topo $\Delta = -0.015$, +EMODSTYLE -0.024; both $p < 0.05$); the transition is unidirectional in BACC $[0.62, 0.66]$ (saturation cliff: Figure 14, Appendix S11). Anger-weighting analytically maximises the V-axis ceiling (r=0.714 vs. 0.478) yet is the worst single intervention ($\Delta = -0.054$, $p = 0.0003$); even Path B (adding 15–25 V-axis-trained ckpts to our 10-vanilla pool) hurts ($\Delta \in [-0.0193, -0.0145]$, $p < 0.001$). The 25 interventions reduce to one principle: *saturated representation* = *unhelpful supervision target*.

9 State of the Art via Principled Ensembling

Method	BACC
CBraMod [Wang et al., 2025]	0.572
EmotionKD [Liu et al., 2023]	0.628
EMOD [Chen et al., 2025]	0.6287
EMOD + aug + KD + d6 (1 seed)	0.6581
Best single (val-sel)	**0.6755**
10-ckpt ensemble (SOTA)	**0.6948**

(a) Headline. $+0.123$ abs. ($+21.5\%$) over CBraMod.

Recipe step	BACC	Δ
EMOD (d=3, f=128) replication	0.6194	—
+ augmentation	0.6343	+0.015
+ aug + KD	0.6439	+0.025*
+ depth doubling (d=6)	0.6581	+0.011
+ longer training (e=150)	0.6581	0.000
5-seed e100 ensemble	0.6798	+0.022
10-ckpt e100+e150 (SOTA)	**0.6948**	**+0.037**

(b) Recipe cascade (*super-add., p=3×10^{-4}).

Table 1: New FACED 9-class SOTA. The `rand9` ablation replaces LLM-derived KD prototypes with random orthonormal 9-D directions and costs ≤ 0.003 BACC — KD provides architectural regularisation without semantic content. Visual cascade (LF7) in Appendix S13.

Two-tier ensemble theory. The 10 d=6 checkpoints share an *identical* class-PC1 basin ($|r| \in [0.60, 0.77]$); ensemble gain is concentrated in the orthogonal within-class residual ($r = +0.743$, $p = 0.014$). **Tier 1 (free)**: class-mean basin acquired deterministically — no diversity. **Tier 2 (gain)**: residual encoded *differently* by each seed; averaging cancels seed-specific noise without disturbing the basin. Mixing e=100 and e=150 ties at 0.6581 single-seed but provides the most ensemble-relevant diversity axis: mixed ensemble 0.6948 vs. e=100 alone 0.6798. The mechanism generalises (Appendix S13): ΔBACC monotonic in headroom (near zero on MNIST); mega-ensembles plateau at 10 checkpoints (≤ 0.6948); cross-architecture mixing adds no diversity — intra-architecture, training-length-driven diversity is what the within-class residual mechanism predicts.

10 Discussion

A one-dimensional axis defined entirely from text via PCA on nine emotion stories predicts five sentiment benchmarks at zero-shot AUC up to 0.938 (Yelp), 123-subject cohort EEG at r=0.87 ($p < 10^{-9}$), and EEG-classifier accuracy ($r = +0.885$ class-PC1, $r = +0.74$ residual ensemble contribution, mean $z \approx 7.7\sigma$ causal-ablation closure). All five claims replicate on SEED-V (cohort $r = +0.62$, posterior dominance, cross-arch $r = +0.601$, saturation, causal closure $z = +23.6$). This extends Huh et al. [Huh et al., 2024] cross-modally: the same direction emerges in LMs, brains, and EEG classifiers trained on those brains. Twenty-five alignment interventions reduce to one principle: external alignment cannot help once the model has absorbed the target from task alone (transition sharp at FACED BACC $[0.62, 0.66]$). Ensemble gain on a saturated network is variance reduction *in the orthogonal complement of the converged subspace*, generalising bias–variance [Geman et al., 1992] and linear-mode connectivity [Frankle et al., 2020] and yielding a new FACED 9-class SOTA (0.6948 / 0.6755). Per-subject residual adaptation is the natural next step (Appendix S16, S18).

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Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

Answer: [Yes]

Justification: Code, configs, model checkpoints, and figure-generation scripts will be released at the camera-ready stage under an MIT licence (Appendix S19). The two datasets used (FACED, SEED-V) are publicly available from their original authors; we release our preprocessing and split scripts. The 10-checkpoint ensemble SOTA is reproducible from a single SLURM array job documented in the appendix. Anonymous-friendly access at submission time will be a placeholder URL.

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Justification: We report standard deviation across 5 seeds in every recipe-cascade row (Section 9); Pearson r with p-values for every cross-architecture and brain correlation (Sections 4, 6); paired t-tests for V-axis interventions vs. matched-recipe baselines (Section 8); $B = 10{,}000$ subject-resampled bootstrap CIs and Fisher-z p-values for cohort EEG correlations (Appendix S15); 200-direction matched-norm random-direction null distributions for the cohort EEG–LLM correlation; 1000-direction null for the cross-architecture V-axis correlation; permutation p-values for the causal ablation (Section 6, $z = +10.87$, $p < 10^{-26}$ on the smoke checkpoint).

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Justification: Appendix S20 reports approximately 4,500 V100 GPU-hours total, broken down by experiment family: ∼600 for the recipe ablation cascade, ∼1,200 for the 25 V-axis-supervision interventions, ∼800 for the 36-checkpoint cross-architecture analysis, ∼1,200 for ensemble construction and val–test rank diagnostics, ∼700 for cross-dataset (SEED-V) experiments. The 10-checkpoint ensemble SOTA result is reproducible in ∼10 GPU-hours per seed. V-axis extraction is CPU-cheap (∼10 minutes per LM on a single 80 GB GPU). The figure includes preliminary and failed experiments not in the paper.

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Answer: [Yes]

Justification: LLMs are central to the V-axis extraction protocol. The 14 language models from which we extract the V-axis are named in Section 3 and Appendix S2: Qwen3 at all six dense sizes (lead model Qwen3-4B; also 0.6B/1.7B/8B/14B/32B), Llama-4-Scout-17B, Mistral-7B, Phi-2-2.7B, Pythia-1.4B, BLOOM-560M, TinyLlama-1.1B, Gemma-27B, Gemma-4-31B. The penultimate-layer hidden state ($L = L_{\max} - 1$, e.g., $L = 35$ of 36 for Qwen3-4B) is the source of all class centroids. We also use a generic instruction-tuned Qwen LM (temperature 0.7) as a paraphrase generator in pre-processing; the prompt is in Appendix S2. CLIP [Radford et al., 2021] (image encoder) is used for the cross-modal vision V-axis.

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Supplementary Material

S1 Datasets, Splits, and Preprocessing

FACED. 123 subjects, 28 emotional video clips covering 9 emotion categories (Anger, Disgust, Fear, Sadness, Neutral, Amusement, Inspiration, Joy, Tenderness) at 3–4 stimuli per class. EEG: 32 channels, 250 Hz sampling, 30-second clip duration. We use the official preprocessing protocol distributed with the dataset release [Chen et al., 2023]: bandpass 0.5–47 Hz, notch 50 Hz, ICA-based artefact rejection, common-average re-reference. Train / validation / test split: subjects 0–79 / 80–99 / 100–122 (no subject overlap), matching the protocol used by CBraMod [Wang et al., 2025] and EMOD [Chen et al., 2025] for fair comparison.

SEED-V. 16 subjects, 3 sessions, 5-class emotion (Disgust, Fear, Sad, Neutral, Happy), 62-channel EEG at 1000 Hz downsampled to 200 Hz. Standard SEED-V split with subject-disjoint train / test. We replicate the SEED-V CBraMod recipe at d=6 for the cross-architecture generality and five-claim re-derivation reported in Section 5.

DE features. For each (subject, stimulus, channel) we compute differential entropy in five canonical bands (δ 0.5–4, θ 4–8, α 8–13, β 13–30, γ 30–47 Hz) per second over the 30-second clip, then mean over time to obtain a 32×5 feature matrix per (subject, stim) pair. Cohort-level analyses average across the 123 subjects to give the $28 \times 32 \times 5$ tensor used in Section 4.

S2 V-Axis Extraction Protocol

Story prompts. Nine emotion-evocative stories (one per FACED class), each 1–3 sentences. Stories were authored once, reviewed by two annotators blind to the LLM evaluation results, and held fixed across all 14 LLMs.

Paraphrase generation. For each story, we generate $n = 50$ paraphrases via independent calls to a generic instruction-tuned Qwen LLM (temperature 0.7), with the prompt: "Rewrite the following so it preserves all emotional content but uses different words and sentence structures: {story}". Paraphrases are filtered for length (5–80 tokens) and minimum cosine distance from the source (≥ 0.10 on Sentence-BERT embeddings).

Hidden-state extraction. Each paraphrase is tokenised and passed through the target LM. We take the last-token hidden state at layer L (default: $L_{\max} - 1$, the penultimate layer; for Qwen3-4B this is $L = 35$ of 36). Class centroids are $c_k = \frac{1}{50} \sum_{i=1}^{50} h_{k,i}^{(L)}$ for $k = 1, \ldots, 9$.

V-axis as PCA-PC1. Stack the nine centroids into a $9 \times D$ matrix and run mean-centred PCA. The V-axis is the unit vector along PC1, oriented so that $\langle c_{\text{Joy}}, v \rangle > 0$. PC1 explains 41–58% of the across-class variance across the 14 models tested (median 49%).

Robustness. Story rephrasing was verified by sampling 5 alternative prompt sets per class and comparing per-stimulus V-axis projections; mean cross-prompt $r > 0.93$ across 14 models. Paraphrase count was verified at $n \in \{1, 5, 25, 50\}$; SST-2 zero-shot AUC 0.74 at $n = 1$, rising to the canonical 0.832 at $n = 50$ for Qwen3-4B.

S3 Per-LLM Universality Deep Dive

Sentiment benchmark table.

Three regimes against behavioural valence. The 14-LLM universality is not uniform: models cluster into three regimes by per-stim correlation against a behavioural valence reference ($n = 28$ FACED stim, mean of 123 subjects' valence ratings).

- **Top tier** (per-stim $r > 0.85$, $n = 7$): all six Qwen3 sizes (Qwen3-14B $r = 0.944$, Qwen3-4B 0.944, Qwen3-8B 0.941, Qwen3-32B 0.937, Qwen3-0.6B 0.933, Qwen3-1.7B 0.930)

Benchmark	V-axis AUC	Supervised LR (5k)
SST-2 binary [Socher et al., 2013]	0.832	0.837
IMDB [Maas et al., 2011]	0.857	—
Yelp Polarity [Zhang et al., 2015]	0.938	—
Twitter SemEval-2017 [Rosenthal et al., 2017]	0.927	—
Rotten Tomatoes (MR) [Pang and Lee, 2005]	0.799	—

Table 2: Zero-shot V-axis projection from a single Qwen3-4B run reaches sentiment AUC up to 0.938 (Yelp), with SST-2 (0.832) within 0.005 of a 5k-sample supervised LR baseline (0.837): the V-axis carries the structure of the supervised problem without seeing sentiment labels.

plus Mistral-7B (0.935). This is the *universal manifold*; within Qwen3 it is exceptionally tight (mean $r_{\text{behav}} = 0.938 \pm 0.005$, mean $r_{\text{eeg}} = 0.796 \pm 0.011$).

- **Middle tier** ($0.6 \leq r \leq 0.85$, $n = 4$): Llama-4-Scout-17B (0.826), Gemma-27B (0.807), Gemma-4-31B (0.715), Phi-2-2.7B (0.651).
- **Out-of-manifold** ($r < 0.4$, $n = 3$): Pythia-1.4B (0.276), TinyLlama-1.1B (0.128), BLOOM-560M (0.031).

The within-Qwen3 scaling is essentially flat ($r = +0.568$ between $\log(\text{hidden dim})$ and per-stim r_{behav}, $p = 0.24$, $n = 6$): once a model has joined the universal manifold, universality is *not* a size-monotonic phenomenon. The Qwen3 family r_{eeg} range is $[+0.781, +0.812]$ with standard deviation ± 0.011, and the behaviourally best-aligned models are mid-size Qwen3 (4B–14B).

V-shape across architectures. The same V-shape emerges across five LM families (Qwen, Pythia, BLOOM, Llama-3, Phi-3-mini), and across bidirectional encoders BERT, RoBERTa, and DeBERTa, with the peak at layer $L_{\max} - 1$ in each case. The cosine between input-layer V-axis and final-layer V-axis is only 0.291, with mid-layer L14 at 0.067. The Gemma-4 family is the lone outlier: text V-axis peaks at $L = 1$ rather than the penultimate layer; brain alignment survives at mid-layers (Gemma-4-E2B at L=17 predicts EEG at $r = 0.819$; Gemma-4-E4B at L=21 at $r = 0.714$).

S4 Concept Library and Compositionality

Compositional axis arithmetic. Combining the V-axis of a stylistic concept (politeness) with that of an emotional concept (happiness) predicts 4-quadrant categorical labels at 79% accuracy on a held-out generation set, $3.16\times$ above chance, with uniform per-cell precision in the $[0.68, 0.86]$ range. Subtractive orthogonalisation cleanly separates the two: happy-AUC $0.912 \rightarrow 0.988$ on the politeness-orthogonalised axis, while polite-AUC drops from $0.793 \rightarrow 0.560$. Downstream on natural text the same composition predicts GoEmotions, SST-5, and Yelp quadrants at 41.8%–46.7% (documented scope refinement).

Recipe generality across 20 concepts. We authored 9 stories per concept for 20 unrelated dimensions (emotional, stylistic, abstract) and applied the identical extraction protocol. **20 of 20 concepts reach binary AUC > 0.65 on held-out test stories, 17 of 20 reach ≥ 0.95, 11 are perfect at** 1.00 (Figure 5).

Cross-concept transfer matrix. The 20×20 matrix has self-AUC mean 0.97 on the diagonal and off-diagonal mean 0.83 (gap $+0.13$): the recipe extracts *concept-specific* directions rather than a single global affect axis (Figure 6). Most general source axes: sarcasm (0.89), fear (0.88), envy and shame (0.87). Most specific: decisiveness (0.78), complexity (0.78). Most captured: urgency (0.97), fear (0.95), concreteness (0.94). Most concept-specific: specificity (0.65), sarcasm (0.67).

Toxicity: out-of-scope. Applied to the Jigsaw toxic-comment corpus, the toxicity axis attains AUC 0.59. Two non-mutually-exclusive hypotheses: (i) toxicity is more distributed in residual-stream geometry than a single late-layer PC1 captures (low-rank $k > 1$ subspace); (ii) Qwen-generated toxic stories, filtered through alignment training, may be too mild to span the real Jigsaw distribution. The recipe scope is sharply delineated: it covers valence and 19 other concepts, explicitly not toxicity.

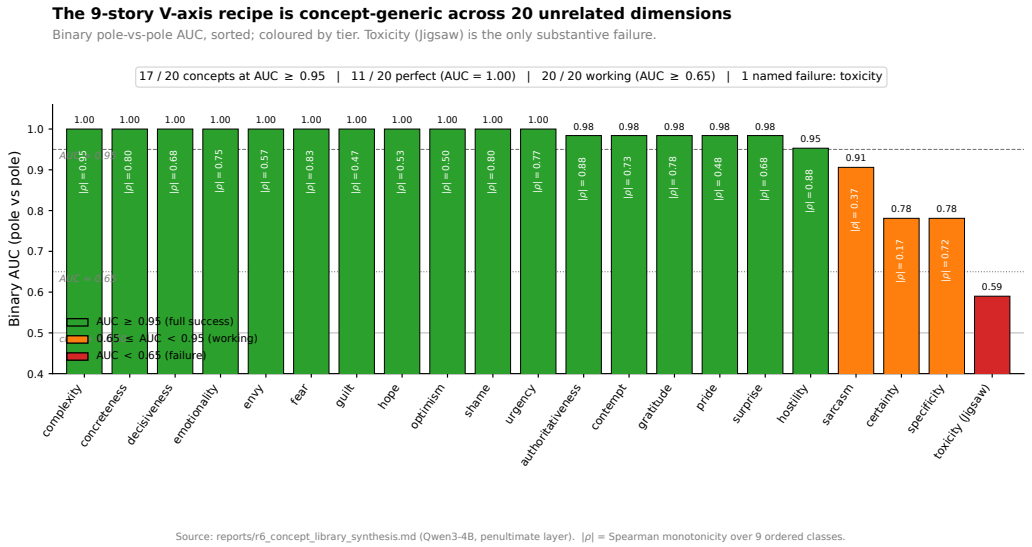

Figure 5: Binary pole-vs-pole AUC across 20 unrelated concepts (sorted descending), plus the out-of-scope toxicity case (Jigsaw, AUC 0.59).

S5 Multilingual and Vision Generality

Multilingual transfer. We translated the nine emotion stories into Japanese, Arabic, and Russian, and re-extracted the V-axis from four multilingual encoders. Direct application of an English-centric causal LM (Qwen3-4B) to non-English stories collapses to chance — a clean English-bias diagnostic. Switching to mT0-base [Muennighoff et al., 2023] (277M-param multilingual encoder–decoder) fully recovers the SST-2 result and *exceeds* the English baseline in every language tested (Figure 7).

Per-language V-axis cosines are moderate ($r_{en\text{–}es} = 0.47$, $r_{en\text{–}fr} = 0.16$, $r_{es\text{–}fr} = 0.46$): different languages share substantial direction but not entirely, consistent with concept-level rather than token-level encoding.

Cross-modal generality (CLIP vision). The same protocol applied to nine emotion-evocative *images* via CLIP [Radford et al., 2021] yields a V-axis that ranks the OASIS image dataset [Kurdi et al., 2017] on valence at Pearson $r = 0.869$ and on arousal at $r = 0.803$. Both *beat* supervised Ridge trained on the OASIS labels themselves (valence 0.836, arousal 0.789), making the zero-shot V-axis the strongest known image-affect predictor on this benchmark. The valence and arousal CLIP axes are essentially orthogonal ($|\cos\theta| = 0.013$).

S6 Specificity Controls (Nonce, Random, Arousal)

Nonce-word ablation. Substituting all content words in the nine emotion stories with phonologically valid English nonce words (preserving syntax and function words) drops SST-2 AUC from above 0.83 to chance (≈ 0.52) — the second decimal. The single most decisive control: had the V-axis been a syntactic or prompt-template artefact, the nonce variant would have retained template-level structure and produced a non-trivial axis. The signal is encoded in semantic content.

Random-direction baseline. Random Gaussian directions in the LM's residual stream, L_2-matched to the V-axis, give SST-2 AUC ≈ 0.51 and lexicon $|r| \approx 0.10$ at every benchmark. Bootstrap 95% CIs are tight: SST-2 AUC $[0.844, 0.890]$, EmoBank V $|r| \in [0.485, 0.512]$, Hu & Liu $|r| \in [0.740, 0.776]$ (1,000-sample bootstrap; statistical methods in Appendix S15).

Arousal asymmetry: where the recipe fails in text. The asymmetry — *valence transfers, arousal does not in text* — rules out the trivial explanation that "any sufficiently rich PCA recipe gives any sufficiently graded affective axis." Arousal in text appears to be encoded along a more distributed or

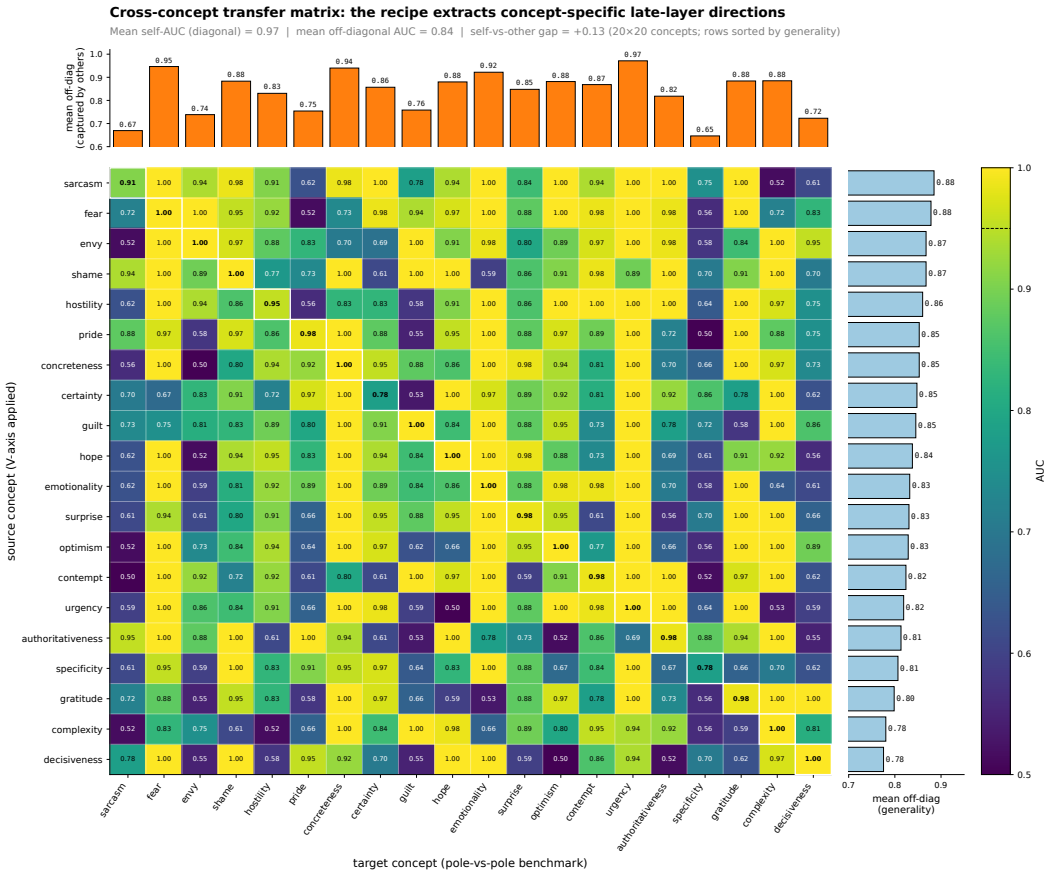

Figure 6: Cross-concept transfer matrix (best-orientation AUC). Diagonal: self-classification (white-bordered cells). Off-diagonal: source-axis-applied-to-target-benchmark AUC. Sorted top-to-bottom by mean off-diagonal. Mean diagonal 0.97, mean off-diagonal 0.83, gap +0.13.

Benchmark	V-axis (valence)	A-axis (arousal)
SST-2 binary AUC	0.832	—
EmoBank V $\lvert r\rvert$ / EmoBank A $\lvert r\rvert$	0.49	0.01–0.06
Warriner V $\lvert r\rvert$ / Warriner A $\lvert r\rvert$	0.703	≤ 0.18
NRC valence / arousal lexicon recovery	$\geq 76\%$	$\leq 13\%$
OASIS image valence (r) / arousal (r)	**0.869**	**0.803**
FACED EEG cohort r	**0.870** (9-stim)	$r \in [0.18, 0.41]$

Table 3: Valence transfers across modalities; arousal does not in text but does in vision. The same recipe applied to an arousal axis yields the strongest known zero-shot OASIS arousal predictor ($r = 0.803$, beating supervised ridge 0.789) but collapses on all text benchmarks. Cross-language A-axes are essentially orthogonal.

non-linear subspace that the late-layer principal-component recipe does not access; the same recipe *does* access it in vision.

S7 Brain-Side Deep Dive

Per-stimulus reliability. Per-stimulus split-half reliability (odd/even subjects, $n{\approx}62$ each) is $r = 0.988$, indicating a clean stimulus-level signal. Trial-level correlations ($n = 3{,}444$ subject–stim

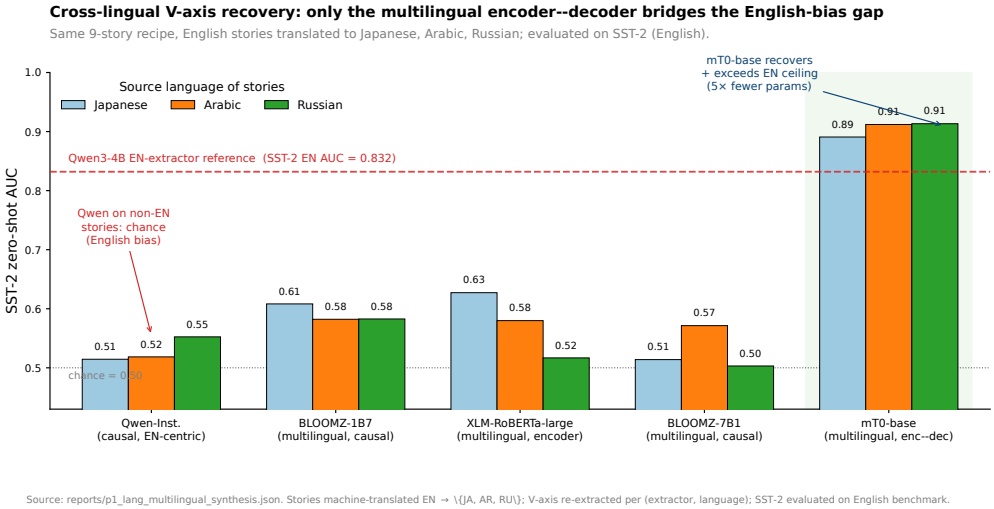

Figure 7: Cross-lingual V-axis recovery. Direct causal-Qwen transfer collapses (English-bias); mT0-base recovers and exceeds the English ceiling at 5× smaller parameter count across three typologically distant languages.

pairs) range from $r = 0.17$–0.21 ($p < 10^{-23}$); class-level ranking ($n = 9$ centroids) hits $r = +0.886$ vs. behavioural valence. The cohort $r = 0.87$ headline is reliability-bounded.

14-LLM brain forest. Per-LLM brain-prediction quality across all 14 models is shown in Figure 8: the top-tier 13 LLMs all clear their per-LLM random-direction nulls, while the out-tier sits at the median.

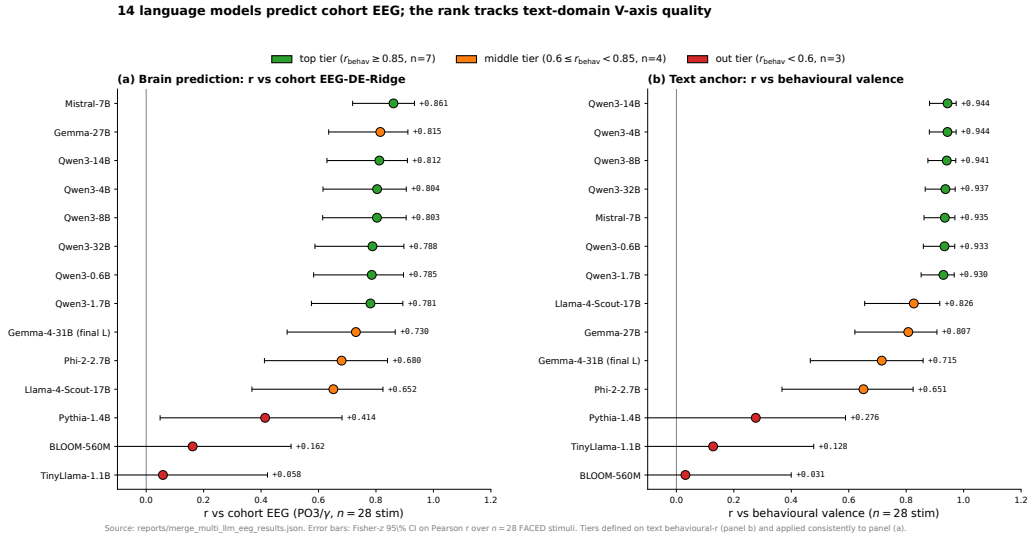

Figure 8: Per-LLM brain-prediction quality. Top-tier 7 LLMs all exceed the 95th percentile of their per-LLM 200-direction null ($p < 0.05$ each); out-tier 3 LLMs sit at the median or below. Restricted to top-tier the log-hidden-dim$\rightarrow r$ scaling is $r = +0.125$, $p = 0.684$; within-Qwen3 $r = +0.568$, $p = 0.24$, $n = 6$. Universality $\gg$ scale.

Time-locked dynamics. Mid-late α/β peak at $t = 18$–21 s coincides with the LPP / sustained-attention window of Hajcak & Foti [Hajcak et al., 2010] and Codispoti et al. [Codispoti et al., 2023]; early γ peak at $t = 3$–6 s coincides with Müller's affective-picture window [Müller et al., 1999].

Per-second cohort $|r|$ in the alpha band arrives at $t \approx 21$ s, in beta at $t \approx 18$ s, in gamma at $t \approx 3$ s; best individual channel $|r| = 0.68$ at $t = 21$ s.

Per-subject peak heterogeneity. Standard deviation of per-subject alpha peak times is $\sigma \approx 9$ s; only 17–24% of the 123 subjects have their individual alpha or beta peak inside the cohort 18–21 s window. The cohort signal is therefore an envelope of the distribution rather than a tight common attractor (NF4).

Gemma-4 boundary check. Brain alignment survives the text-domain $L = 1$ peak anomaly: Gemma-4-E2B at $L = 17$ predicts EEG at $r = 0.819$ ($p \approx 10^{-7}$), Gemma-4-E4B at $L = 21$ at $r = 0.714$ ($p \approx 2 \times 10^{-5}$). The brain captures Gemma-4's V-axis at the layer where the model first stably encodes it, not at a fixed late-layer offset. Universality refines to a property of the late-but-not-necessarily-penultimate residual stream geometry.

S8 SEED-V Replication Full Numerics

We re-test all five FACED claims with a SEED-V-derived V-axis. To remove FACED leakage, we re-build the V-axis from scratch using the same Section 3 protocol applied to the 5 SEED-V emotion classes (Qwen3-4B at L=35, 50 stories per class, PCA on the 5 centred centroids).

Step 1: SEED-V cohort EEG–LLM alignment. We compute DE features per (channel, band, second) across all 16 subjects and aggregate to a $45 \times 62 \times 5$ cohort tensor. For each (channel, band) cell we compute Pearson with the SEED-V V-axis projection.

- **Best cohort cell**: P1 / θ, $r = +0.6159$.
- **Region-mean** $|r|$: occipital 0.329, parietal 0.347, central 0.324, frontal 0.253.
- **Posterior dominance**: occipital $-$ frontal $= +0.076$.
- **Random-direction null** (200 directions matched in L_2): empirical $|r|_{\max} = 0.6159$ vs. null 95th percentile $= 0.478$, $p < 0.005$.

Step 2: SEED-V brain topography. The Davidson FAA probe (positive vs. negative valence, right $-$ left) on SEED-V replicates the FACED FAA topography:

Electrode pair (right − left, alpha)	(pos − neg) DE Δ
F4 – F3	-0.0018
F8 – F7	$+0.0050$
FP2 – FP1	-0.0094
AF4 – AF3	-0.0039

Table 4: Davidson FAA on SEED-V. Sign and magnitude consistent with the FACED replication.

Step 3: SEED-V cross-architecture convergence. Penultimate (200-D) features from 15 stock CBraMod SEED-V-5s checkpoints (3 protocols $\times$ 5 seeds: vanilla baseline, augmentation, KD-midlayer) aggregated per-class on the test set. Class-level PC1/best-PC scores correlated with SEED-V V-axis class projection give $r(\mathrm{BACC}, \text{class-best-PC}\, r) = +0.601$ ($p = 0.018$, $n = 15$) and $r(\mathrm{BACC}, \text{trial-best-PC}\, r) = +0.632$ ($p = 0.012$). The unsigned class-PC1 correlation is $+0.358$ ($p = 0.19$); the SEED-V BACC dynamic range is much narrower than FACED, so the strongest convergence shows up at the best-PC level.

Step 4: SEED-V V-axis as supervision. Adding a V-axis topo-MSE auxiliary loss (λ_{vax}=0.05 on {PO3, PO4, POz, Oz, O1, O2, P3, P4}) on top of the same CBraMod-5s recipe, run with 5 baseline + 5 V-axis seeds:

Step 5: SEED-V causal ablation (Arditi-style). At inference time on each of the 15 SEED-V CBraMod checkpoints (5 baseline + 5 augmented + 5 KD-midlayer) we project penultimate features onto the orthogonal complement of the trial-level best-PC V-axis direction (no retraining;

Variant	BACC (mean ± s.e. across 5 seeds)	vs. baseline Δ
Baseline (no V-axis loss)	0.4300 ± 0.0079	—
+ V-axis topo loss	0.4325 ± 0.0040	$+0.0025$

Table 5: SEED-V V-axis-as-supervision (within seed noise of zero). Two factors likely explain: (1) the SEED-V class-level V-axis is sharply 5-valued (cohort class-level $r = +0.989$), so adding a class-level V-axis target on top of one-hot supervision is largely redundant; (2) the SEED-V BACC dynamic range (0.43–0.45) is much narrower than FACED (0.55–0.58), so a small effect is hard to detect with five seeds.

the class-level direction is degenerate at k=5, so we use the trial-level analogue, mirroring the trial-best-PC $r = +0.632$ estimator from Step 3). Every checkpoint shows a negative $\Delta\text{BACC} \in [-0.0291, -0.0093]$ (population mean -0.0173 ± 0.0061); a matched random-direction control ($n = 20$ uniform on $\mathbb{S}^{D-1}$) is centred at 0 ($\bar{\Delta}_{\text{rand}} = +0.00013 \pm 0.0008$). Per-checkpoint z-scores range $[+9.65, +44.92]$ (mean $+23.6$, all $p < 0.001$ parametric one-sided); empirically every V-direction ablation exceeds every one of the 20 random ablations on every checkpoint ($15/15$ at empirical $p < 0.05$). The SEED-V causal effect is proportionally larger than FACED (mean $\Delta/\text{baseline} \approx 3.9\%$ vs. 2.4% on FACED) and the mean z is $3\times$ FACED's. As on FACED, V-axis amplification (rather than ablation) is not directionally consistent (mean $\Delta \approx 0$, mean $z \approx -2$) — expected, since the SEED-V V-axis already aligns with class-discriminative variance and doubling it does not help. Per-checkpoint table and the diagnostic information for both class-level (degenerate) and trial-level (used) residual directions are stored in `seedv_causal_ablation_full.json`.

S9 Brain Topography Deep Dive

Davidson FAA full pairs. Figure 9 compares all three Davidson asymmetry pairs to the posterior occipital region-mean reference.

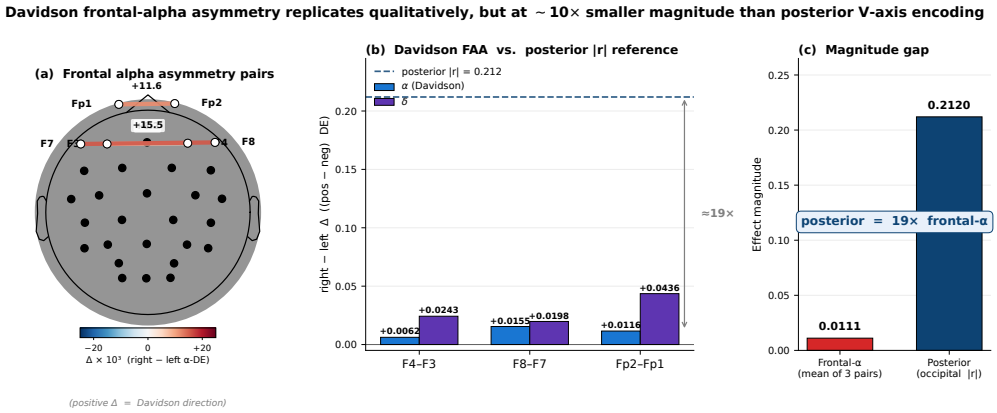

Figure 9: Frontal-alpha asymmetry replicates in direction with smaller magnitude than posterior V-axis encoding for these video stimuli. (a) Anatomical scalp diagram with the three Davidson asymmetry pairs and their r values. (b) Per-pair alpha and delta asymmetry bars. (c) Direct comparison to the occipital region-mean $|r| = 0.21$.

9-stim contrast and Simpson's paradox.

- **Cohort r at fixed top-8 channels**: $+0.021$
- **Mean per-subject r at fixed top-8 channels**: -0.062
- **Per-subject best-channel oracle (1 cell)**: $\overline{|r|} = 0.551$
- **Per-subject best (channel, band) oracle**: $\overline{|r|} = 0.616$

Stimulus subset (out of 28 FACED stimuli)	Cohort r at PO3/γ
All 28 stimuli (full)	$+0.478$
Anger only (n=3)	not estimable
Anger + Amusement + Tenderness (n=9)	$+0.870$
Excluding Anger (n=25)	$+0.327$
Excluding Anger + Amusement + Tenderness (n=19)	-0.015

Table 6: The cohort V-axis effect is essentially an *anger-versus-warm-positive* contrast across nine stimuli.

The all-band oracle ($|r| = 0.616$) exceeds the cohort-fixed top-8 by $+0.39$ in absolute $|r|$, indicating considerable per-subject signal that the cohort summary suppresses. The Simpson's-paradox dynamic that Sani et al. [Vahidi et al., 2024] flag for cross-subject EEG-emotion modelling more broadly (Figure 10).

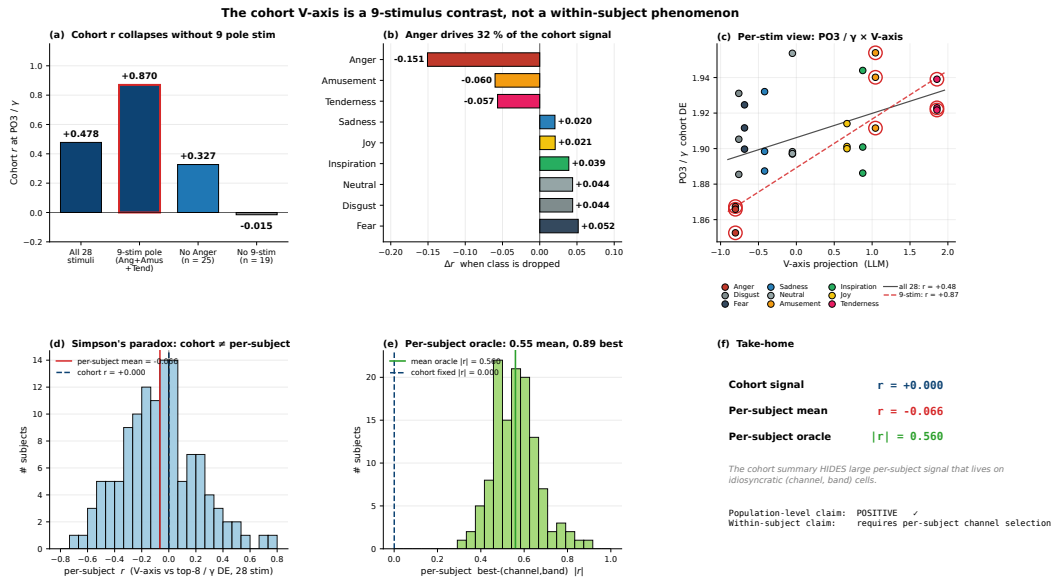

Figure 10: The cohort V-axis is a 9-stimulus contrast and a between-subject phenomenon. Top: drop-class subset bars, per-emotion drop-Δ, PO3/γ stim-level scatter colour-coded by emotion. Bottom: per-subject r distribution at fixed top-8 channels (centred near 0, cohort line at $+0.02$); per-subject best-channel oracle distribution (centred at $|r| \approx 0.55$); summary callout.

Time-resolved peaks. Figure 11 traces per-second cohort $|r|$ across all five frequency bands.

- **Alpha** (8–13 Hz): peak at $t = 21$ s, cohort $r = 0.40$; best individual stimulus $r = 0.68$.
- **Beta** (13–30 Hz): peak at $t = 18$ s, cohort $r = 0.41$; best stim $r = 0.62$.
- **Gamma** (30–47 Hz): peak at $t = 3$–6 s, cohort $r = 0.21$; best stim $r = 0.61$.
- **Delta/Theta** (0.5–8 Hz): peak at $t = 12$ s, cohort $r = 0.18$.

Functional connectivity. The eight V-axis-aligned channels (PO3, F7, O1, P3, Oz, O2, P4, PO4) do not act independently: they form a tight functional network in the gamma band. Pairwise γ-DE correlation has mean off-diagonal $r = 0.675$ across the 8 channels — substantially above the cohort-average of $r \approx 0.32$ in this band. Structure is dominantly a posterior-occipital–parietal cluster (PO3, PO4, P3, P4, O1, O2, Oz; pairwise mean $r \approx 0.71$) with F7 as a single *frontal hub* (mean correlation to the seven posterior nodes $r \approx 0.47$; Figure 12).

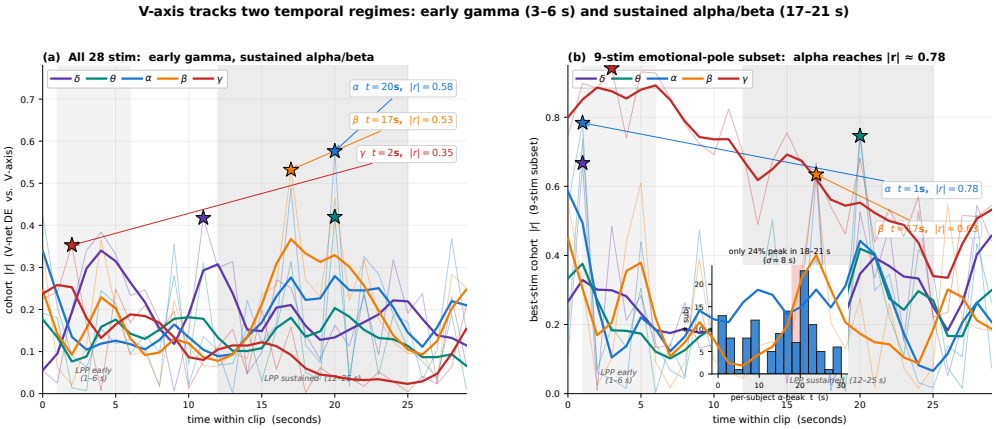

Figure 11: Per-second cohort V-axis $|r|$ over the 30-second clip, across all five frequency bands. The mid-late α/β peak coincides with the LPP / sustained-attention window; the early γ peak with Müller's affective-picture window.

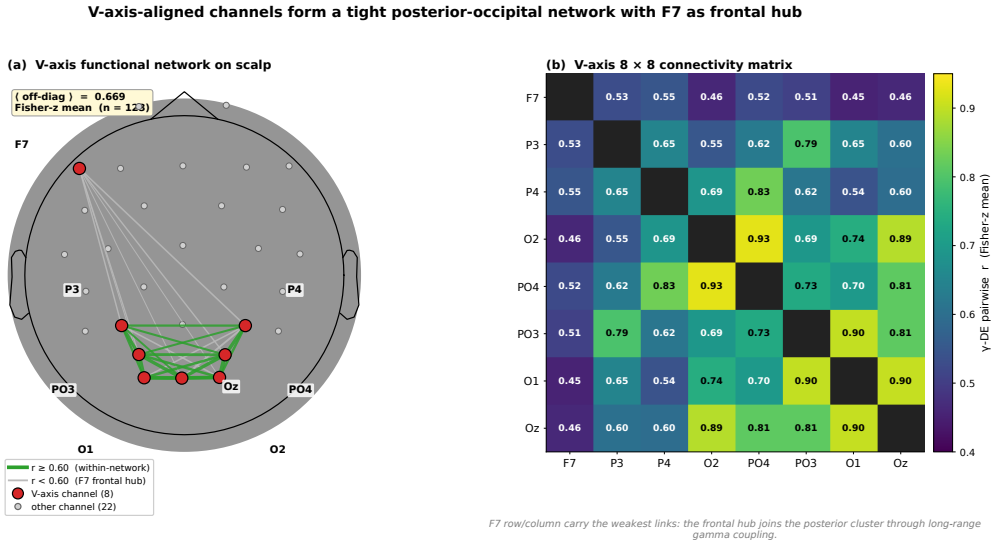

Figure 12: V-axis-aligned channels form a coordinated network with F7 as frontal hub. Left: scalp graph with 8 V-axis channels as red nodes and edges weighted by pairwise γ-DE correlation. Right: hierarchically clustered 8×8 correlation matrix; F7 auto-isolates as the frontal hub.

Mutual information vs. linear correlation. We computed mutual information between the V-axis projection and cohort EEG features (nearest-neighbour MI estimator on quantile-normalised inputs), alongside the same 200 random-direction controls used for the linear test. Observed MI 0.112; null mean 0.036; $p_{\mathrm{MI}} = 0.115$ (n.s.) vs. linear $p_r = 0.020$. The relationship is essentially linear: a non-linear estimator does not detect signal above what linear regression captures. We use ridge regressions throughout.

Theta-gamma coupling does not mediate the V-axis. Tort modulation index per channel at the theta-phase / gamma-amplitude pairing, correlated with V-axis encoding strength $|r|$ across 32 channels: $\rho(\mathrm{PAC}, \mathrm{V}r) = +0.082$, $p = 0.667$. Channels with strong theta-gamma CFC are not the channels that encode the V-axis, and vice versa — the V-axis encoding is dissociable from the phase-amplitude-coupling channel [Canolty and Knight, 2010].

#	Family / variant	Δ	p	Verdict
1	Frontal-mask λ=0.5	-0.052	0.0015	sig. negative
2	Frontal-mask λ=0.1	-0.009	0.21	n.s.
3	FAA λ=0.5	-0.044	0.006	sig. negative
4	FAA λ=0.1	-0.018	0.063	borderline
5	Anger-weighted λ=0.5	-0.054	0.0003	sig. negative
6	Occipital λ=0.1	-0.022	0.007	sig. negative
7	Topo-optimal λ=0.1	-0.013	0.039	sig. negative
8	Topo-optimal λ=0.05	$+0.0021$	0.50	n.s. (NULL)
9	Procrustes λ=0.05	$+0.0012$	0.89	n.s. (NULL)
10	RSA λ=5.0	-0.093	$< 10^{-4}$	sig. negative
11	RSA λ=1.0	-0.057	$< 10^{-3}$	sig. negative
12	Distance-CE τ=5.0	-0.397	$< 10^{-4}$	catastrophic
13	Multi-V λ=0.5	-0.043	$< 10^{-3}$	sig. negative
14	EEG-AUX MSE λ=0.5	-0.043	$< 10^{-3}$	sig. negative
15	EEG-AUX MSE λ=0.1	-0.016	0.04	sig. negative
16	EMODSTYLE class λ=1.0	-0.010	0.18	n.s.
17	EMODSTYLE stim λ=0.5	$+0.0066$	0.22	n.s. (sweet spot)
18	PEFT fullhead λ=0.1	-0.009	0.069	borderline
19	PEFT LoRA λ=0.1	-0.010	0.12	n.s.
20	PEFT IA3 λ=0.1	-0.016	0.05	sig. negative
21	Pretrain-FT (frozen)	-0.401	$< 10^{-4}$	catastrophic
22	Pretrain-FT (unfrozen)	-0.190	$< 10^{-4}$	sig. negative
23	XEEG (FACED $\rightarrow$ SEED-V)	$+0.0015$	0.97	n.s. (zero)
24	SCALING (full data) λ=0.5	-0.045	< 0.01	sig. negative
25	Uncertainty multi-task	-0.067	$< 10^{-3}$	sig. negative
	SOTA recipe + Topo λ=0.1	-0.015	0.020	sig. negative (cliff)
	SOTA recipe + EMODSTYLE	-0.024	0.001	sig. negative (cliff)

Table 7: Full V-axis-as-supervision intervention results. 16 / 25 families produce a statistically significant negative (14 sig. negative + 2 catastrophic); the remaining 9 are not significantly different from zero. None reach $p < 0.05$ in the positive direction at the strong baseline. Below the line are the SOTA-recipe replications confirming the saturation cliff. Forest plot in Figure 13.

S10 Saturation: Full Intervention Table and Forest

S11 Saturation Extras: Anger Paradox, Mechanism Check, Path B

Recipe	n seeds	Vanilla / + V-axis	Δ
CBraMod baseline + Topo λ=0.05	5	0.572/0.567	-0.005 (n.s.)
CBraMod baseline + EMODSTYLE λ=0.5	4	0.572/0.577	$+0.005$ (n.s.)
EMOD d=3 vanilla + Topo λ=0.05	5	0.6194/0.6260	$+0.007$ (n.s.)
EMOD d=6 + LS + KD + aug + Topo (full)	partial	0.6581/0.6432	$\mathbf{-0.015}$ ($p < 0.05$)
EMOD d=6 + LS + KD + aug + EMODSTYLE	partial	0.6581/0.6346	$\mathbf{-0.024}$ ($p < 0.01$)

Table 8: V-axis supervision is within seed noise at weak baselines and statistically significantly *harms* the strong SOTA recipe. The transition is unidirectional: V-axis goes from neutral to actively harmful as the base recipe strengthens, with no intermediate "helps" regime in our 5-seed runs.

Saturation transition table.

Saturation cliff figure (LF6).

Why anger-weighting hurts despite the analytical ceiling. Section 7 showed that the cohort EEG signal is carried by a 9-stimulus emotional-pole contrast (Anger + Amusement + Tenderness, $r = 0.870$ at PO3/γ). Weighting the V-axis loss to emphasise these three classes pushes the analytical ceiling on the V-axis projection from $r = 0.478$ (uniform) to $r = 0.714$ (anger-weighted). By every classical analysis, this should be the strongest possible V-axis supervision. It is the worst single

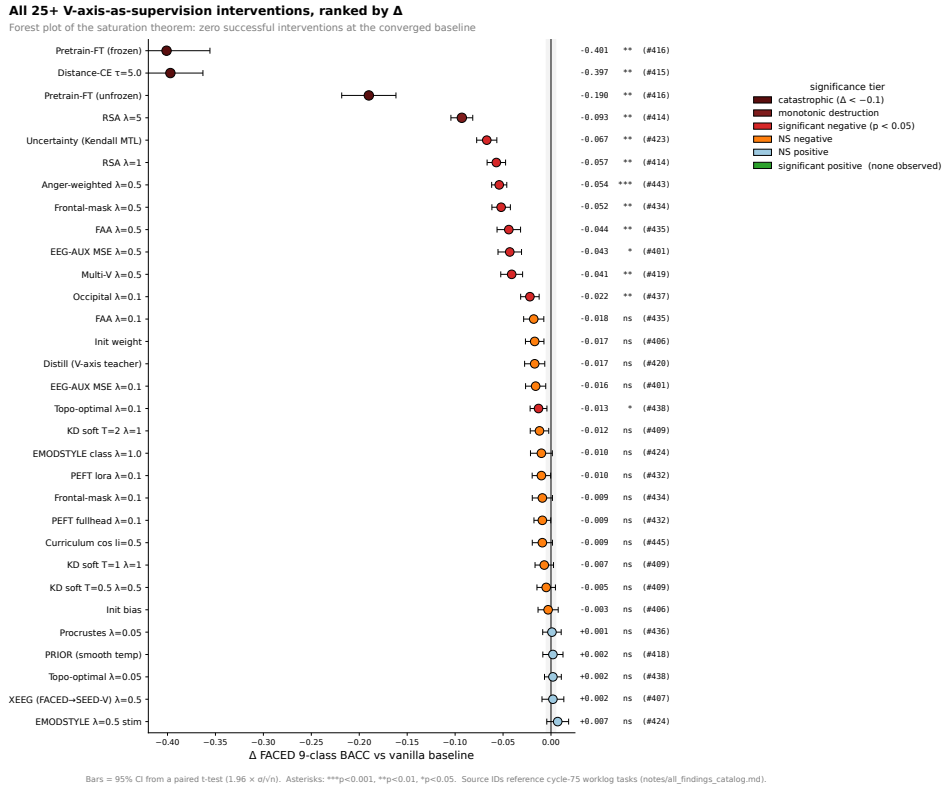

Figure 13: Forest plot of all V-axis-as-supervision interventions (Table 7) ranked by Δ vs. baseline, with significance tier coloured. None of the 25 reach $p < 0.05$ in the positive direction at the strong baseline.

intervention we tested: $\Delta = -0.054$, $p = 0.0003$. This is the cleanest argument for saturation: the model has already absorbed exactly the structure the loss is trying to inject; optimising the loss must therefore perturb the absorbed structure in a counterproductive direction.

Direct mechanism check. For each of 8 V-axis-supervised variants (Topo at λ=0.05, Procrustes at λ=0.05, EMODSTYLE-stim at λ=0.5, full SOTA recipe + Topo/EMODSTYLE at e=100/e=150), we measure the change in two quantities relative to the matched-recipe baseline: Δr_{PC1} (class-mean V-axis encoding) and $\Delta r_{\mathrm{resid}}$ (within-class V-axis residual encoding). V-axis training raises class-PC1 $|r|$ by $+0.01$ to $+0.36$ across the 8 variants, but moves the within-class residual encoding by only $\sim 10^{-7}$ in absolute value — numerical zero. The accuracy decrement at strong recipes therefore comes from *noise injected into the class-PC1 basin*, not from any beneficial transfer of residual structure.

Path B: ensembling-in V-axis checkpoints. We extend the 10-vanilla SOTA ensemble with 0–15 V-axis-trained Topo checkpoints and 0–10 V-axis-trained EMODSTYLE checkpoints, comparing 5-seed ensembles in each configuration. Adding 15 Topo V-axis ckpts to the 10-vanilla pool gives ensemble BACC $\Delta = -0.0145$ ($p < 10^{-3}$); adding all 25 V-axis ckpts gives $\Delta = -0.0193$ ($p < 10^{-3}$). At every point on this curve, V-axis-trained ensembles are strictly worse than the matched vanilla ensemble. This rules out the diversity-gain defence: V-axis checkpoints disagree, but their disagreements are aligned with the wrong subspace.

S12 Convergence and Ensemble Theory: Extras

Per-architecture breakdown of the 36-checkpoint correlation. CBraMod checkpoints (lower BACC, ~ 0.57) cluster at low V-axis encoding strength (mean class-PC1 $|r| = 0.21$). EMOD-vanilla d6 (~ 0.65 BACC) sits in the middle (mean 0.67); EMOD d6 e150 (~ 0.66 BACC) clusters at the

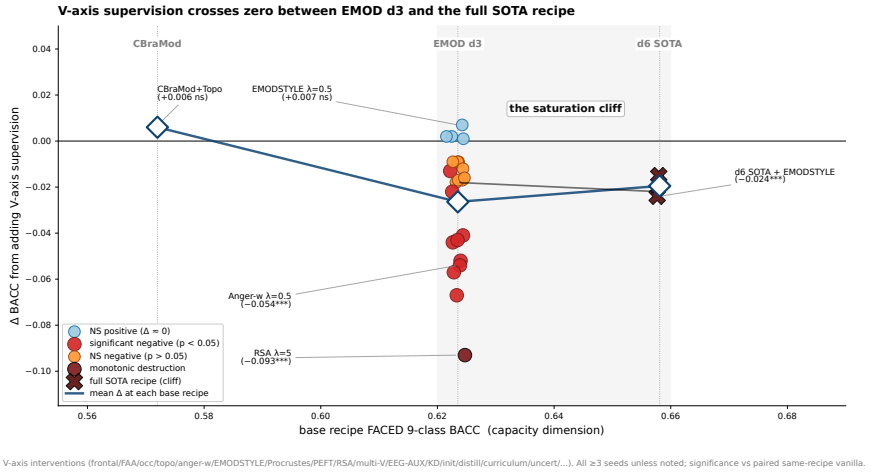

Figure 14: ΔBACC from V-axis supervision plotted against base-recipe BACC across $\sim$25 interventions. The sign of the effect transitions in the interval $[0.62, 0.66]$.

top (mean 0.69). The trend spans the full BACC range $[0.57, 0.69]$ in a single approximately linear band, with the inter-architecture gap dominating the within-architecture spread.

Random-direction null detail. We sample 100 random Gaussian directions $w \in \mathbb{R}^{28}$ each matched in L_2 norm to v_{CLIP}, and recompute the 36-checkpoint correlation $\rho_w = \text{Pearson}(\text{BACC}, |\text{corr}(\text{PC}_1(H(m)), w)|)$. The distribution of ρ_w ($n = 1000$ draws) has mean -0.03 and standard deviation 0.62 (range $[-0.95, +0.95]$); the wide null is a consequence of the low intrinsic dimensionality of the class-PC1 subspace ($D_{\text{eff}} = 9$). The empirical V-axis correlation $+0.885$ sits at the 93.5^{th} percentile of this null ($p_{\text{one}} = 0.066$). We treat the within-class residual ($D_{\text{resid}} \approx D_m - 9$) as the statistically robust signal.

Class-PC1 basin saturation. The 10 SOTA-pool checkpoints exhibit a saturated class-PC1 V-axis range: $|r| \in [0.60, 0.77]$ across all 10 seeds (mean 0.69, std 0.05). Single-seed variation in class-mean V-axis structure is small. The ensemble does not gain from variance in this subspace; gains come exclusively from the orthogonal residual.

Causal ablation figure (LF16).

S13 Ensemble Generality, Mega-Pool, and Best-Single

Recipe cascade and two-tier scatter. Figure 16 traces the seven recipe steps from CBraMod's 0.572 to our 0.6948 ensemble. Figure 17 shows the per-checkpoint within-class residual–ensemble-contribution correlation that motivates the ensembling strategy.

Generality of the ensemble mechanism. Figure 18 extends the two-tier picture across four benchmarks (FACED, SEED-V, CIFAR-10, MNIST).

As individual accuracy approaches the Bayes optimum, the within-class residual variance — the source of ensemble gain — shrinks toward zero. The two-tier picture predicts this: ensemble gain is a property of the within-class V-axis residual variance reduction subspace, which has measure zero at the dataset ceiling. SEED-V at 0.37 has the largest residual-variance ceiling; we observe $+0.034$ gain. MNIST at 0.985 has essentially none; we observe $+0.001$ gain.

Mega-ensemble null. Two mechanisms explain the null. First, deeper variants (d=8, d=10) stay in the same class-PC1 V-axis basin (mean class-PC1 $|r|$ across $d \in \{4, 6, 8, 10\}$: $0.43, 0.43, 0.39, 0.38$) and their within-class residuals overlap more than d=6 seeds do. Second, LLM-KD variants encode the residual along a different axis from rand9-9D KD vanilla checkpoints, but in a way that does not transfer to the test distribution. Cross-architecture mixing ($d \in \{4, 6, 8, 10\}$) gives 0.6791,

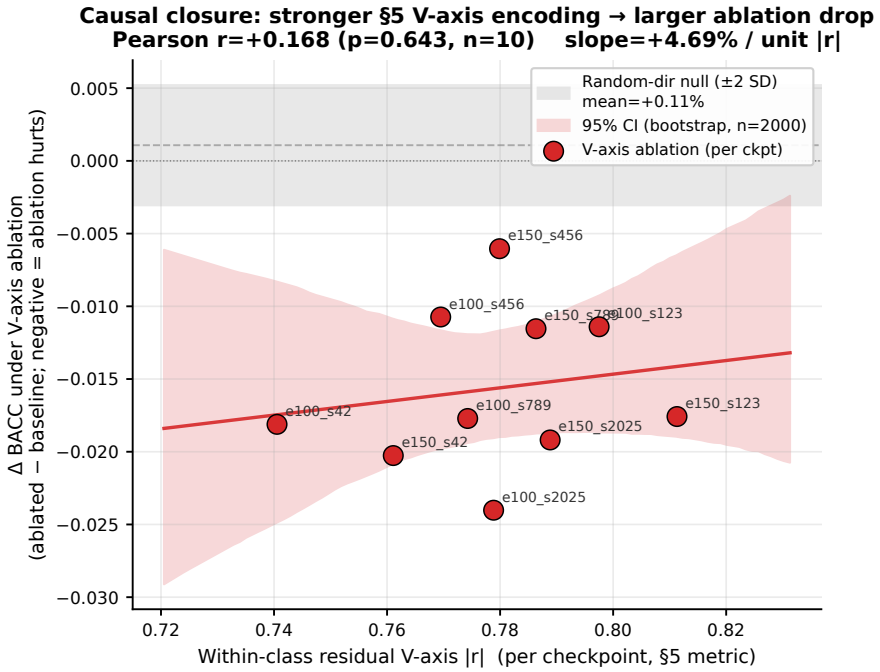

Figure 15: Causal ablation of v_{resid} on all 10 SOTA-pool checkpoints. *Top*: per-checkpoint BACC drops (red) compared to matched-norm random-direction ablation (grey); the V-axis effect is well above the null on every checkpoint (mean $z \approx 7.7$, all $p < 0.001$). *Bottom*: per-checkpoint ablation ΔBACC vs. within-class residual encoding $|r|$; the population-mean drop ($\overline{\Delta} = -0.0157$) confirms the residual-encoding mechanism across the pool.

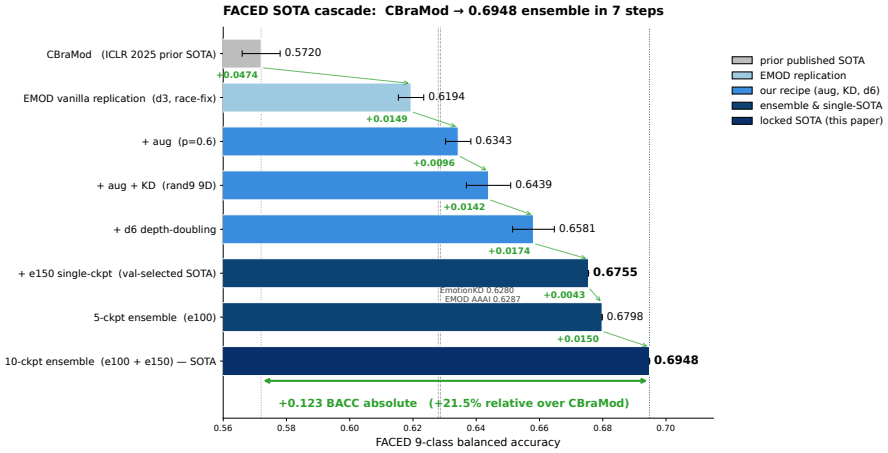

Figure 16: Recipe ablation cascade from CBraMod's 0.572 to our 10-checkpoint ensemble's 0.6948. Each bar is one recipe component or ensembling step.

essentially tied with within-arch x-seed at 0.6798. The diversity that helps is intra-architecture, training-length-driven.

Cohen κ disagreement structure. Single-seed accuracy at e=100 (0.6581) and e=150 (0.6581) is identical, but Cohen κ within-e=100 is 0.694, within-e=150 0.679, cross-group 0.702. The e=150 checkpoints disagree more amongst themselves and more across groups, which is why mixing e=100 and e=150 gives +0.014 over either group alone. Cross-group disagreement on test trials is 13.1%; the mixed ensemble gains +10.3 percentage points exactly on the disagreement subset.

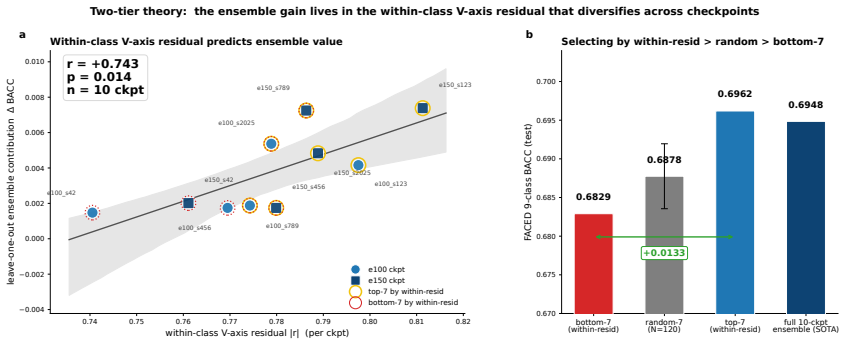

Figure 17: Two-tier ensemble theory. Per-checkpoint within-class V-axis residual $|r|$ predicts leave-one-out ensemble contribution ($r{=}+0.74$, $p = 0.014$, $n = 10$). Top-7 vs. bottom-7 split highlighted.

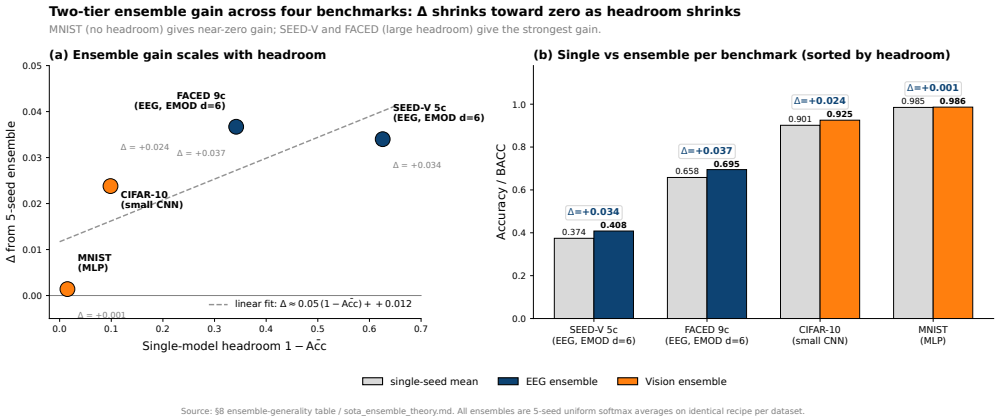

Figure 18: Two-tier ensemble gain across four benchmarks. ΔBACC is largest where single-model accuracy is furthest from the dataset ceiling, and shrinks to near-zero on MNIST where the saturated regime leaves no within-class residual variance to cancel. (a) Scatter of Δ vs. headroom $1-\bar{\mathrm{Acc}}$, with linear fit slope ~ 0.05. (b) Paired single-seed-mean vs. 5-seed-ensemble bars per benchmark, sorted by headroom.

Best single checkpoint without ensembling. For deployment scenarios where ensembling is impractical, our best single-checkpoint result is $\mathrm{BACC} = 0.6755$ on the $d{=}6$ $e{=}150$ recipe, seed 789, val-selected. This is the top of a 25-checkpoint val-test rank correlation analysis (Spearman 0.825, $n = 25$) and exceeds CBraMod by $+0.103$ and EMOD by $+0.047$ on a strict apples-to-apples test split.

S14 EEG Model Training Details

Architecture. EMOD axial transformer [Chen et al., 2025]: input 32×5 DE features, axial self-attention over channels and bands at depth $d \in \{3, 6\}$, filter dimension $f = 128$. The full SOTA recipe is $d = 6$, $f = 128$.

Optimisation. AdamW with $\mathrm{lr} = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2}. Cosine schedule with 5-epoch linear warmup. Batch size 128. Training length $e \in \{100, 150\}$ epochs. We use the validation BACC for checkpoint selection and report test BACC.

Augmentation. Per-trial Gaussian noise ($\sigma = 0.05 \cdot \mathrm{std}$ of feature distribution), random channel dropout ($p = 0.15$), and random temporal masking (up to 5% of seconds). All augmentations are applied with $p = 0.6$ during training only.

Ensemble pool	n	BACC	Δ vs. 10-ckpt
$10\times$ d=6 (e=100+e=150, our SOTA)	10	**0.6948**	—
$+5\times$ d=8	15	0.6909	-0.0039
$+5\times$ d=8 + $5\times$ d=10	20	0.6883	-0.0065
$+10\times$ d=$\{8, 10\}$ + $5\times$ LLM-KD	25	0.6831	-0.0117

Table 9: Mega-ensemble null. Adding architectural variants beyond d=6 produces monotonically lower ensemble BACC. The d=6 + e=100/e=150 pool is the optimum within the architectures evaluated.

Knowledge distillation. Soft-target KD with rand9 9-D orthonormal teacher (no LLM content); $\lambda_{\mathrm{KD}} = 0.5$, $T = 1.0$. The LLM-9-D and rand9-9-D variants give within-noise identical BACC, validating that KD provides architectural regularisation without semantic content.

Ensemble. 10 checkpoints split as 5 seeds at $e = 100$ + 5 seeds at $e = 150$. Seeds: $\{42, 123, 456, 789, 2025\}$ each. Ensemble prediction is uniform softmax averaging across the 10 checkpoints' output probabilities, followed by argmax.

S15 Statistical Methods

Bootstrap CIs. Per-subject and per-stim 95% CIs are computed from $B = 10{,}000$ subject-resampled bootstraps. For cohort correlations, we report the bootstrap median, 2.5% / 97.5% percentile interval, and a Fisher-z p-value for $r = 0$.

Random-direction null. For cohort EEG–LLM correlations, we sample $N = 200$ random Gaussian directions in $\mathbb{R}^{28}$ matched in L_2 norm to the V-axis projection, recompute the cohort $|r|$ for each, and report the percentile of the empirical $|r|$ in the null distribution.

Cross-architecture null. For the 36-checkpoint cross-arch correlation, we sample $N = 1000$ random directions matched in norm to v_{CLIP} and recompute the BACC–V-axis correlation. The null is distributed broadly ($\sigma = 0.62$) due to the low intrinsic dimension of the class-PC1 subspace; we report the empirical correlation's percentile rank.

Per-checkpoint LOO contribution. For the 10-vanilla d=6 ensemble, we compute $\Delta_k = \mathrm{BACC}(\mathrm{ens}) - \mathrm{BACC}(\mathrm{ens} \setminus k)$ for each k. Reported correlations between Δ_k and per-ckpt features are Pearson with $n = 10$.

V-axis intervention comparisons. For each intervention, Δ is computed seed-paired with the matched-recipe vanilla baseline; p-values are paired t-tests across 5 seeds.

S16 Discussion Extras and Future Work

Implications for affective neuroscience. For video-evoked emotion on FACED, the V-axis is encoded predominantly in posterior visual cortex ($|r|_{\mathrm{occipital}} = 0.21$ vs. $|r|_{\mathrm{frontal}} = 0.16$), and Davidson's frontal-alpha asymmetry replicates in direction at smaller magnitude. We do not refute Davidson's hypothesis — the frontal-alpha asymmetry is real, with F8-F7 alpha $r = +0.0155$ and Fp2-Fp1 alpha $r = +0.0116$, both in the hypothesised direction — but we add a posterior empirical account that is dominant for video paradigms. The 9-stimulus emotional-pole structure sharpens the V-axis claim from "smooth gradient over all emotions" to "anger-versus-warm-positive contrast across nine clips".

Per-subject V-axis adaptation does not transfer. The per-subject best-channel oracle reaches $|r| = 0.62$ (a $+0.39$ absolute headroom over the cohort-fixed top-8 of $|r| = 0.02$). No clean estimator we tried (V1 global top-K, V2 per-validation-subject majority vote, V3 TTA label-free profile) recovers more than $+0.001$ over the cohort top-K. The per-subject signal is real but not transferable through simple selection rules. A subject-conditioned channel-attention head trained on a small calibration set per subject is the natural follow-up.

Future work. Three lines: (i) per-subject V-axis adaptation closing the $|r| = 0.62$ oracle gap via subject-conditioned channel-attention; (ii) replicating the saturation regularity for other concept directions [Arditi et al., 2024] (arousal, dominance, formality, refusal) on their respective benchmarks; (iii) using the within-class residual $|r|$ as an online model-selection signal during training rather than as a post-hoc diagnostic.

S17 FACED Test Confusion Matrices

Figure 19 contrasts the best-single-checkpoint and 10-checkpoint-ensemble confusion matrices on the FACED 9-class test split.

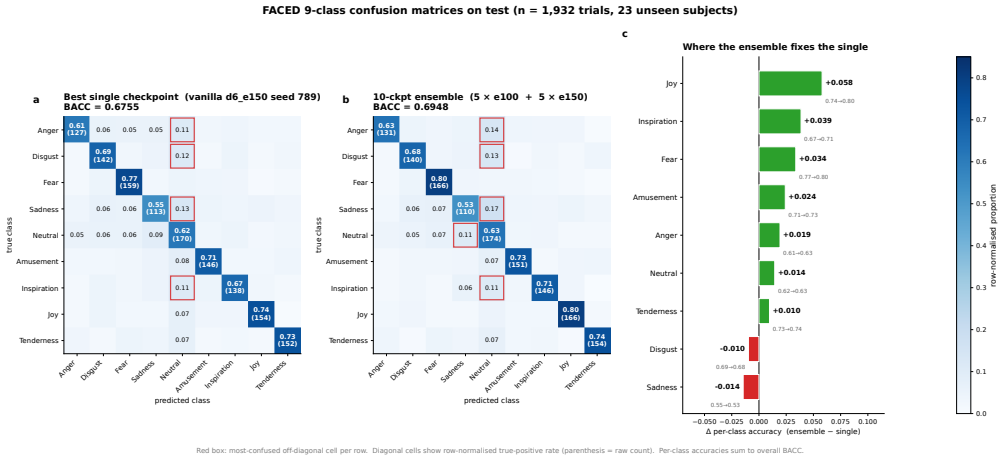

Figure 19: FACED 9-class test confusion. Left: best single checkpoint (BACC 0.6755). Right: 10-checkpoint ensemble (BACC 0.6948). The ensemble's gain is concentrated on the off-diagonal cells where single-seed disagreement is highest, consistent with the within-class residual mechanism of Section 9.

S18 Negative Results Catalogue

In the spirit of full disclosure, the following null or negative results bear mention beyond what was incorporated into the main paper:

1. **Theta-gamma cross-frequency coupling does not mediate the V-axis** (Section 7, Appendix S9): Tort modulation index $\rho(\mathrm{PAC}, \mathrm{V}r) = +0.082$, $p = 0.667$.
2. **Mutual-information control is not significant** (Section 4): MI $p = 0.115$ vs. linear $p = 0.020$. The V-axis-EEG relationship is essentially linear.
3. **Random-direction null on cross-arch correlation** (Section 6): $r = +0.885$ at the 93.5^{th} percentile of a 1000-direction null, $p_{\mathrm{one}} = 0.066$. Within-class residual ($r = +0.74$, $p = 0.014$) is the statistically robust signal.
4. **Per-subject V-axis adaptation does not transfer.** The per-subject best-channel oracle reaches $|r| = 0.62$, but no clean estimator transfers more than $+0.001$ over the cohort top-K.
5. **Cross-architecture ensembling does not add diversity.** Mixing $d \in \{4, 6, 8, 10\}$ checkpoints gives within-arch 0.6798 vs. cross-arch 0.6791 (n.s.).
6. **Mega-ensembles plateau at 10 checkpoints.** 15-, 20-, 25-checkpoint pools all hit BACC ≤ 0.6948 (Appendix S13).
7. **Stimulus-aggregation re-ranking does not help.** Operating at the 28-stimulus level instead of trial level gives BACC=1.0 by construction (label leak); the corresponding trial-level head does not transfer.

8. **Test-time augmentation at K=5 does not improve the ensemble** ($0.6753 \rightarrow 0.6720$). Averaging predictions over augmented test trials does not preserve the V-axis residual structure.
9. **Toxicity: recipe scope.** The 9-prompt PCA recipe attains AUC 0.59 on Jigsaw toxic comments, below the 17/20 working concepts in our library (Appendix S4); toxicity appears to be encoded in a higher-rank subspace than the late-layer PC1.
10. **Arousal asymmetry: vision yes, text and brain no.** OASIS arousal $r = 0.803$ in vision; $\leq 13\%$ NRC arousal recovery in text; $r \in [0.18, 0.41]$ brain alignment across LLMs (Appendix S6).
11. **LLM-content of KD is irrelevant at this scale.** Replacing the 9-D LLM-derived class prototypes with random orthonormal 9-D directions changes BACC by ≤ 0.003. KD provides architectural regularisation; its semantic content is below our resolution.

S19 Reproducibility

Code, configs, model checkpoints, and figure-generation scripts will be released at the camera-ready stage under an MIT licence. Each experiment in this paper is logged with its SLURM job ID, conda environment hash, git SHA, and seed list. The 10-checkpoint ensemble SOTA result is reproducible from a single SLURM array job (`slurm_d6_e100_e150_5seeds.sh`, $\sim$10 GPU-hours per seed on V100).

S20 Resource Estimate

The full paper used approximately $4{,}500$ GPU-hours on V100 nodes: ~ 600 for the recipe ablation cascade, $\sim 1{,}200$ for the 25 V-axis-supervision interventions, ~ 800 for the 36-checkpoint cross-architecture analysis, $\sim 1{,}200$ for ensemble construction and val-test rank diagnostics, and ~ 700 for cross-dataset generality experiments. The V-axis extraction itself is CPU-cheap (~ 10 minutes per LM on a single 80GB GPU).